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A Meta-Analysis of the Effects of Group-Based Interventions on Social Reintegration of Marginalized Adults with Mental Illness

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Abstract

This preregistered systematic review and meta-analysis examines group-based community interventions for socially marginalized adults with mental illness in OECD countries. Searches identified 28,980 records; 62 studies were included in the review and 49 studies, contributing 349 effect sizes, in the meta-analysis. Outcomes were classified as social reintegration, including social functioning, loneliness, employment, and quality of life, or mental health, including anxiety, depression, and psychotic symptoms. Multivariate meta-analysis showed that interventions improved social reintegration (g = 0.195, 95% CI[0.122, 0.268]) and mental health (g = 0.215, 95% CI[0.090, 0.340]) relative to controls. These averages masked substantial heterogeneity: effects varied widely across studies, and predictive distributions of true effects ranged from near zero to moderate in both domains. Measured moderators explained little of this variation, indicating that important sources of effect variation remain unidentified. Effects in the two domains were positively associated, with studies showing larger social reintegration effects also tending to show larger mental health effects. Results were robust across sensitivity analyses and showed little indication of publication bias. Nearly half of studies were preregistered and contributed approximately two-thirds of all effect sizes, strengthening confidence in the evidence. Preregistered studies yielded smaller social reintegration effects than non-preregistered studies, but these effects remained statistically significant, substantively meaningful, and 1.6 times the typical improvement under treatment as usual. Together, the findings suggest that group-based interventions can support multidimensional recovery while potentially offering a cost-effective alternative to individual care.

Significance Statement

Mental illness is a leading contributor to disability, social exclusion, and economic costs worldwide. Many affected individuals face multiple challenges, including loneliness, homelessness, unemployment, and substance misuse, yet effective and scalable solutions remain scarce. By synthesizing evidence from 49 studies, we show that group-based community interventions improve both mental health and social reintegration outcomes for marginalized adults with mental illness. Beyond symptom reduction, group-based approaches may help rebuild social connections and community participation. Our findings suggest that investing in group-based interventions could improve outcomes for some of society's most vulnerable citizens while increasing the efficiency and reach of overstretched mental health services.

Introduction

Mental illness is often accompanied by social and personal adversities that extend beyond psychiatric symptoms. Adults with mental illness are at elevated risk of substance use, homelessness, unemployment, poverty, self-harm, criminal justice involvement, and social isolation. These co-occurring problems can intensify stigma, deepen exclusion, and increase public costs.[1–4] Mental illness and marginalization may also become mutually reinforcing: people may withdraw from social participation because of anticipated or experienced stigma, while isolation and exclusion can further worsen mental health.[5–7] For many adults, the challenge is therefore not only to manage symptoms, but also to sustain relationships, participate in social life, and avoid persistent exclusion.

This broader reality poses a challenge for intervention research. Many mental health interventions are designed primarily to reduce symptoms of a specific disorder. Yet, symptom reduction alone does not capture whether people become more socially integrated or are able to live meaningful and connected lives. For adults facing both psychiatric difficulties and social marginalization, recovery may also depend on housing stability, employment, reduced substance use, stronger social ties, improved quality of life, and a greater sense of agency.

A range of interventions has been developed to address these needs, including occupational therapy, case management, psychoeducation, supportive psychotherapy, and mentoring. However, such interventions are often resource-intensive, and evidence of their effectiveness remains mixed.[8–10] At the same time, mental healthcare systems in many high-income countries have shifted from hospital-centered care toward community-based and outpatient services. This transition has increased pressure on local systems to deliver effective support under constrained budgets.[11] Within this context, group-based interventions have become increasingly attractive because they can serve multiple participants simultaneously and may offer a more scalable alternative to individually delivered care with low costs.[12,13]

Group-based interventions may also offer benefits beyond efficiency. Their distinctive feature is that interpersonal contact and social support are built into the intervention itself. Group-based interventions will often be provided in a small, selected group of individuals who meet regularly with a therapist or case worker.[14] This may be especially important for populations whose difficulties are closely linked to loneliness, stigma, and impaired social functioning[12]. Across mental health conditions, interpersonal functioning and social support are important predictors of relapse, recurrence, and longer-term adjustment.[15–18] These are also modifiable factors relevant to sustained recovery[19]. Group settings may therefore address mechanisms that individual treatment cannot fully target, including social learning, mutual recognition, shared norms, and supportive relationships.[12,17–20]

These potential benefits can be understood through recovery[21] and social identity[22] perspectives. A recovery perspective emphasizes that improvement is not limited to symptom remission, but includes building a satisfying and meaningful life despite ongoing difficulties. Group-based interventions may support recovery by fostering coping skills, confidence, hope, and participation in social and occupational roles. A social identity perspective suggests that becoming part of a group may reshape how marginalized individuals see themselves and others. Group membership can foster a sense of belonging, validate experiences, and promote health-related behaviors through shared norms and mutual support.

Yet the promise of group-based interventions should not be assumed. Group processes can also be harmful[14,23,24]. Poor cohesion, breaches of confidentiality, invalidation or rejection by other members may reinforce feelings of isolation and low self-worth. More generally, not all participants are equally likely to benefit from the same treatment format. Depending on diagnosis, comorbidity, interpersonal style, or broader circumstances, some individuals may benefit less from group-based care than from individual intervention. The key question is therefore not simply whether group-based interventions work, but for whom, under what conditions, and whether they improve outcomes relevant to both mental health and social reintegration among marginalized adults with mental illness.

Existing evidence does not provide a clear answer regarding the effects of group-based interventions on social reintegration outcomes. Reviews of psychiatric group interventions[25,26] have largely focused on specific diagnoses and typically prioritize symptom reduction as the main endpoint. Other reviews have examined broader outcomes in areas such as psychosis recovery, substance use, homelessness, and trauma, but these studies tend to focus on narrower populations, single comorbidities, or specific intervention models[27–32]. As a result, the evidence base remains fragmented. We still know relatively little about whether group-based community interventions improve the wider outcomes that matter for adults living with both mental illness and social marginalization.

This gap is important for science and policy. The social and economic burden of comorbidity is substantial. Yet, many evaluations rely on narrow clinical outcomes that may underestimate the value of interventions affecting housing, work, social functioning, or quality of life. Policymakers are increasingly asked to fund group-based community interventions because of their potential scalability and lower cost relative to individual treatment. These decisions require stronger evidence.

The present review addresses this gap by systematically evaluating group-based community interventions for adults with mental illness who also experience social or personal problems associated with marginalization. We examine effects not only on mental health but also on social reintegration outcomes, including substance use, homelessness, employment, loneliness, well-being, and quality of life. By synthesizing evidence across intervention types and populations, the review assesses whether group-based community interventions more generally support multidimensional recovery in a population for whom both clinical and social outcomes are central.

We address four research questions. First, what is the overall effect of group-based community interventions on social reintegration outcomes, such as problem behavior, subjective well-being, homelessness, poverty, and employment? Second, do effects on social reintegration vary across substantive (i.e., theoretically and empirically relevant) or methodological moderators? Third, what is the overall effect of these interventions on mental health outcomes? Fourth, do effects on mental health vary across substantive or methodological moderators?

These questions follow the analysis plan specified in our preregistered and peer-reviewed protocol[33], although we refined the wording to state the review aims more precisely. Our primary focus is on social reintegration outcomes, which reflect the main aim of the review. Because the protocol primarily focused, albeit indirectly, on social reintegration rather than mental health, analyses of mental health outcomes should be considered exploratory. All materials supporting this manuscript are available on OSF at https://osf.io/s2j9a. We used generative AI for text editing and coding support; we take full responsibility for the final content of the manuscript.

Results

Preregistration has increased over the last decade

As shown in Figure 1, the search identified 62 eligible studies, comprising 51 randomized controlled trials (RCTs) and 11 non-randomized studies with at least one intervention and one control group. Together, these studies included 62 independent samples of participants. Of the 62 eligible studies, 49 studies containing 349 effect sizes were included in the meta-analysis. The remaining 13 studies were excluded from quantitative synthesis because one study was judged to be at critical risk of bias, and 12 studies did not report sufficient information to calculate effect sizes. In the SI Appendix, we assess whether the conclusions of these 12 studies differed from the overall meta-analytic findings, and found no indication that they reached different conclusions.

<insert Figure 1 here>

Figure 2 shows the publication year and preregistration status of the studies included in the meta-analysis. The evidence base has grown substantially since 2009, with 40 (82%) studies published from 2009 onwards. Preregistration also became increasingly common after 2012: among studies published between 2012 and 2022, 23 of 34 studies were preregistered. This pattern suggests a marked improvement in the transparency and methodological quality of the literature over time.

<Insert Figure 2 here>

Diverse types of group-based interventions and participants

The included studies evaluated a broad range of group-based interventions, as highlighted in Table 1.

<Insert Table 1 here>[34–45], [46–54], [55–62], [63–68],[41,69–73],[74–78], [83–85],[86,87],[88,89],[87,90],[91,92],[93,94],[63],[95]

Interventions were categorized by their primary content and therapeutic focus. Most combined symptom management, psychoeducation, social-skills training, or CBT-based techniques reflecting the dual clinical and social aims of the review. Smaller groups of studies addressed addiction, employment, residential support, social network development, positive thinking, trauma-related coping, art therapy, or reading-based reflection. Some interventions included elements from multiple categories, but each study was assigned to the category that best captured its main component.

The included studies covered a clinically diverse population, but with a clear concentration around schizophrenia or other primary psychotic disorders, followed by depressive, bipolar, and anxiety disorders. Indicators of social marginalization were more unevenly reported. Substance abuse was the most frequently specified social or personal problem, whereas homelessness, loneliness, and criminal justice involvement were rarely represented. Notably, nearly half of the studies did not clearly specify the form of social marginalization. Here, 'not clearly specified' means that the study authors did not explicitly state the relevant characteristic, but that it could be identified from their description of the participant sample. Find a full description of the included participants by primary type of mental disorder and primary indicator of social marginalization in the SI Appendix Table S1

Descriptive statistics

Table 3 presents descriptive percentages for the studies and outcomes included in the meta-analysis of social reintegration outcomes. Descriptive averages for the social reintegration outcomes and the descriptive statistics for mental health outcomes can be found in the SI Appendix Tables S2-S4. The meta-analysis of social reintegration outcomes included 46 studies, 205 effect sizes, and 5,406 participants. All studies were journal articles and reported short-term effects, defined per-protocol as outcomes measured within one year after the intervention. Measurement timing within this period did not explain variation in effect sizes.

Studies were conducted mainly in the United States (13 studies), Europe (15), and Commonwealth countries (16), with only two studies from Asia. The most frequently assessed social reintegration outcomes were well-being and quality of life (25 studies), social functioning (17), hope, empowerment, and self-efficacy (12), and alcohol or drug use (8). Less common outcomes included self-esteem, loneliness, physical health, employment, and psychiatric hospitalization; the latter three were combined into an "Other" category for subgroup analyses.

Most outcomes were self-reported: 39 studies contributed 168 self-reported effect sizes, while 12 studies contributed 36 clinician-rated effect sizes. Twenty-four studies reported treatment-on-the-treated effects, and 22 reported intention-to-treat effects.

Participants were, on average, 41 years old (median = 41; range = 25-67), and the samples were 47% male (median = 46; range = 0-81). The average raw sample size was 118 participants, although the median was 60 (range = 16-631), indicating that a few large studies increased the mean. Treatment and control group sizes were generally balanced, and the average treatment group size of the group-based interventions was 7.9 participants (median 8, range 4-22).

Most studies were randomized controlled trials (38 of 46), of which 22 were preregistered. The remaining eight were quasi-experimental. Only three studies accounted for clustering introduced by group-based delivery. The most common comparison condition was treatment as usual, with or without a waiting list (39 studies). Interventions lasted, on average, 18 weeks (median = 12; range = 4-78), with most delivered at a frequency of 0.5 to 2 sessions per week. See Online Appendix B Figure 75-79 for a visualization of the joint distribution of duration and intensity.

<Insert Table 2 here>

The overall descriptive statistical patterns were proportionally similar between social reintegration and mental health outcomes. The mental health dataset included 144 effect sizes derived from 42 studies, encompassing a total of 4679 individual participants. Specifically, nine studies contributed 14 effect sizes for anxiety, 20 studies contributed 37 effect sizes for depression, 29 studies contributed 72 effect sizes for general mental health, and seven studies contributed 21 effect sizes for psychotic symptoms. Of note, three studies[53,57,62] were included in this data only, as they did not report on any usable social reintegration outcomes. In total, 39 studies reported outcomes for both social reintegration and mental health.

The quality of the evidence is generally sound

We conducted comprehensive risk-of-bias assessments to exclude critically flawed evidence, examine whether study quality influenced the meta-analytic findings, and describe the overall quality of the literature. Risk of bias in randomized studies was assessed using Cochrane's RoB 2[96] with results summarized by preregistration status for social reintegration and mental health outcomes; corresponding plots are provided in SI Appendix, Figure S7A. This preregistration distinction was relevant only for randomized trials, as no non-randomized studies were preregistered. Additional risk-of-bias plots are provided in Online Appendix B, Figures 1–24.

Overall, most effect sizes from randomized trials were judged to have low risk of bias or some concerns: approximately 90% for both social reintegration and mental health outcomes. Preregistered studies were generally at lower risk of bias than non-preregistered studies, particularly because prespecified protocols reduced concerns about selective reporting.

Risk of bias was generally higher among non-randomized studies assessed with Cochrane's ROBINS-I[97], as shown in SI Appendix, Figure S7B. Most effect sizes were judged to be at serious risk of bias, primarily because of confounding and deviation from the intended intervention. Other concerns included missing data. Because none of the non-randomized studies had a prespecified protocol, none could receive an overall low risk of bias judgment. We excluded one non-randomized study[74] from the meta-analysis because it was judged to be at critical risk of bias.

Given these concerns, risk of bias was included as a control variable in the covariate-adjusted moderator analyses. However, we found no systematic or substantial differences in effect sizes between studies judged to be at high or serious risk of bias and those judged to be at low or moderate/some concern risk.

Significant impact on both social integration and mental health

Figure 3 depicts the distribution of all effect sizes across studies and the type of outcome. For more detailed forest plots, including study weights, see SI Appendix Figures S2-S3. Moreover, all codes behind the main analysis can be found in the Online Appendix C.

Across the main analyses, group-based community interventions produced statistically significant benefits for both social reintegration and mental health. The meta-analysis of social reintegration yielded a positive overall average effect of 0.195 SD (95% CI [0.122, 0.268], $t(24.9) = 5.51$, $p < 0.001$). This indicates that participants receiving group-based interventions had better social reintegration outcomes than those receiving individual treatment, treatment as usual, or wait-list controls. The effect was approximately 2.3 times larger than the typical improvement observed under treatment as usual.

For mental health outcomes, the meta-analysis yielded a positive overall average effect of 0.215 SD (95% CI [0.090, 0.340], $t(37.5) = 3.48$, $p = .001$). This effect was approximately 1.4 times larger than the typical improvement observed under treatment as usual.

<Insert Figure 3 here>

In addition, and as depicted in Figure 4, we find that effects on social reintegration and mental health also covaried: studies showing larger reintegration gains tended to show larger mental health gains. This suggests that group-based community interventions may support multidimensional recovery, improving both social participation and clinical well-being.

<Insert Figure 4 here>

Substantial heterogeneity across both types of outcomes

There was substantial heterogeneity in both outcome domains, indicating that the effects of group-based community interventions varied considerably across studies and effect sizes. For social reintegration outcomes, heterogeneity was high with I^2 = 88.9% and total heterogeneity of 0.204 SD. Most of this variation occurred within studies rather than between studies, suggesting that effects differed meaningfully across outcome measures within the same study. The 67% prediction interval[1] ranged from −0.012 to 0.401, crossing zero at its lower bound: although the average effect is positive and significant, the predictive distribution implies that a non-trivial minority of future studies could yield null or negative effects, with approximately 82% expected to fall above zero.

Heterogeneity was even larger for mental health outcomes, with I^2 = 99.94% and total heterogeneity of 0.333 SD. In contrast to the reintegration analysis, most variation occurred between studies, suggesting that study-level differences, such as population, intervention type, or implementation context, may be especially important for mental health effects. The 67% prediction interval[2] ranged from −0.120 to 0.549, again crossing zero: roughly 74% of future studies would be expected to show positive effects, but the wider interval indicates that null or negative effects are correspondingly more plausible here than for social reintegration.

Overall, the predictive distributions indicate that group-based interventions are likely to produce positive effects in most future studies, particularly for social reintegration. However, the wider prediction interval for mental health suggests that effects on clinical outcomes are less stable and may depend more strongly on study context, population, or intervention characteristics. However, it is important to remember that effects equivalent to individual treatment may have practical value if more people can be treated with the same resources.

Robust results across sensitivity analyses

Sensitivity analyses showed that the main findings were robust. The overall effect for social reintegration was largely unchanged across alternative assumptions about within-study effect-size correlations, while the mental health effect decreased only slightly as the assumed correlation among within-study effect sizes increased. Leave-one-study-out analyses indicated that no single study drove the overall effects in either outcome domain.

Results were also stable across alternative effect-size calculations and inclusion criteria. When analyses were restricted to low-risk-of-bias effect sizes, estimates were no longer statistically significant, but their magnitude remained close to the main effects, suggesting reduced statistical power rather than a substantive change in conclusions. Overall, the sensitivity analyses support the robustness of the main findings. Find detailed sensitivity results in the Online Appendix D and visualizations of the main sensitivity analyses in the SI Appendix Figures S4-S5.

No indication of publication bias and related issues

We conducted a range of complementary publication bias analyses, as recommended in meta-analysis[98–101], finding little evidence that the main findings were driven by selective publication, selective reporting, or *p*-hacking. As shown in the funnel plots in Figure 5, effect sizes showed no consistent association with standard errors, and publication-bias-adjusted estimates generally

[1] The 95% prediction interval ranged from -0.233 to 0.623 for social reintegration.

[2] The 95% prediction interval ranged from -0.472 to 0.901 for mental health.

remained positive. This pattern may partly reflect the high proportion of preregistered studies, which contributed approximately two-thirds of all effect sizes.

We found some indication of selective reporting among non-preregistered social reintegration effects: negatively signed effects, defined as those with a one-sided *p*-value > 0.50, were four times less likely to be observed than positive but statistically nonsignificant effects (p = 0.02, 95% CI [1.17, 19.23]). However, adjusting for this selection changed the point estimate by less than 0.02 SDs, and we found no corresponding evidence of selection among mental health outcomes. Overall, publication bias therefore does not appear to be a major concern. All publication-bias-adjusted overall average effects are reported in the SI Appendix Table S5, and all other publication-bias tests and visualizations are provided in Online Appendix E.

<Insert Figure 5 here>[102]

Measured moderators explained little heterogeneity, but preregistered studies yielded significantly smaller effects

Given the substantial heterogeneity in both outcome domains, we conducted protocol-informed meta-regression analyses to assess whether effects varied by substantive or methodological moderators. Analyses were conducted separately for social reintegration and mental health outcomes, using both unadjusted and covariate-adjusted models. Because relative moderator differences were generally similar across model specifications, we primarily interpret the unadjusted moderator effects and use the covariate-adjusted estimates to assess the robustness of these results. Full results from all moderator analyses are reported in Tables S6–S11 in the SI Appendix.

Overall, moderators explained rather little of the heterogeneity in both social reintegration and mental health outcomes. Most subgroup estimates clustered around the main effects of the given outcome, and between-subgroup differences were generally small and statistically insignificant. We found no clear evidence that effects differed by outcome type, schizophrenia status, risk of bias, intervention duration, session frequency, follow-up timing, or sample gender composition. Many of the subgroup estimates that departed from this overall pattern were often based on few studies and effect sizes and should therefore be interpreted cautiously.

The main exception to this pattern was preregistration status: preregistered studies produced a substantially smaller effect on social reintegration than non-preregistered studies in both unadjusted and covariate-adjusted models. However, the average effect on social reintegration outcomes among preregistered studies remained statistically significant and substantively meaningful in size (g = 0.130, 95% *CI*[0.030, 0.220]), corresponding to an average effect 1.6 times larger than the typical average improvement in social reintegration under treatment as usual. Notably, however, preregistration explained only a small share of the heterogeneity in effects, leaving substantial variation to be explained by other predictors. For mental health outcomes, the difference between preregistered and non-preregistered studies was not statistically significant, although the direction was similar, with larger effects in non-preregistered studies.

Two further, less pronounced patterns concerned estimand type and intervention type. Across outcome domains and model specifications, ITT estimates were consistently smaller than TOT estimates, although evidence for these differences was strongest in the unadjusted models. The difference reached the 0.10 significance level for social reintegration outcomes and the 0.05 level for mental health outcomes. Nevertheless, for social reintegration outcomes, the ITT estimate remained substantively meaningful in size; for mental health outcomes, it was smaller and closer to the typical improvement under treatment as usual. We also found that CBT-based interventions yielded effects approximately 0.1 SD larger than other intervention types across outcome domains and model specifications.

Control condition also showed a consistent association with effect size. Across outcome domains and model specifications, studies using waitlist controls yielded substantially larger effects than studies using treatment-as-usual or individual-treatment control conditions. In the few studies that used an individual version of the group intervention as the control condition, effects were close to zero, suggesting comparable effectiveness. This finding may still be practically meaningful if group-based interventions achieve similar outcomes at lower cost.

Some moderator patterns differed across outcome domains. Most notably, test type showed opposite associations: clinically rated measures yielded smaller effects for social reintegration outcomes but larger effects for mental health outcomes. For mental health outcomes, participant age was also statistically associated with effect size. However, this association explained only a limited share of the heterogeneity and should be interpreted cautiously because the age range of included samples was relatively narrow (IQR = 39.5–43.6), limiting substantive interpretation and potential explanation of the pattern.

Taken together, the moderator analyses provided sparse evidence that the effectiveness of group-based interventions depends strongly on measured study, sample, or intervention characteristics. Positive average effects were broadly consistent across subgroups, but the remaining heterogeneity across all moderator models indicates that important sources of variation remain unmeasured and may operate both within and between studies. Future research should therefore focus on explaining this heterogeneity. In particular, future studies should examine why participant age, CBT interventions, and preregistration status appear to be associated with differential effects across outcome domains.

Discussion

This review provides robust evidence that group-based community interventions benefit adults experiencing both mental illness and social marginalization, adding to existing evidence base showing the efficacy of group-based interventions[13,25,103–106]. Across heterogeneous studies varying in intervention type, setting, implementation, design, outcomes, and participant characteristics, on average, group-based interventions produced statistically significant improvements in both social reintegration and mental health. Average effects on social reintegration were approximately 2.3 times larger than the typical gains observed under usual individual treatment, while average effects on mental health were approximately 1.4 times larger. Importantly, effects across the two domains covaried: studies showing larger gains in social reintegration also tended to show larger gains in mental health. This suggests that group-based interventions may support multidimensional recovery rather than producing isolated improvements in a single outcome domain.

While the average effects were positive in both domains, the effects were also highly heterogeneous (total heterogeneity of 0.204 SD for social reintegration and 0.333 SD for mental health), and the predictive distributions of true effects included some negative effects as well as some moderately large positive effects. The central finding is therefore not a single effect but a distribution of effects: group-based interventions improve outcomes on average, yet the magnitude of benefit varies substantially from one study and outcome to the next, and a non-trivial minority of contexts or outcomes may yield little or no benefit.

We conducted comprehensive moderator analyses to examine whether this variation in effects could be explained by outcome type, participant characteristics, intervention type, test type, preregistration status, research design, control condition, risk of bias, age, sex composition, intensity, duration, and measurement timing. Overall, moderators explained little of the observed heterogeneity. Close to all heterogeneity remained unexplained, indicating that important sources of variation may be unmeasured.

Against this general pattern of limited moderator explanatory power, preregistration status stood out. Although preregistration explained only a small share of the heterogeneity, preregistered studies yielded significantly smaller effects than non-preregistered studies. At the same time, the only indications of selective reporting were found among non-preregistered social reintegration effects. This pattern suggests that preregistration may partly distinguish a more conservative and less selectively reported evidence base, while also underscoring that the positive overall effects were not driven by non-preregistered studies alone. All types of effects in preregistered studies remained substantively meaningful, and adjustment for selection among non-preregistered studies did not materially change the overall conclusions.

Several limitations should be noted. Though we used comprehensive database searches, grey-literature searches, and hand searches, some potentially relevant references could not be obtained in full text. In addition, though we were able to use highly sophisticated meta-analysis methods, 12 eligible studies could not be included in the meta-analysis because effect sizes could not be calculated. However, supplementary analyses found no indication that these studies reached different conclusions from the quantitative synthesis. Finally, although publication-bias tests were consistently reassuring, such tests are imperfect and cannot fully exclude selective reporting. Yet, the high number of preregistered studies might mitigate this issue.

The evidence appears broadly applicable to adults with psychiatric diagnoses who also experience personal or social difficulties. The included studies covered a wide range of countries, diagnoses, interventions, and outcomes, and we found no clear evidence that effects differed systematically by sample type, intervention type, or national context. However, the review focused specifically on adults experiencing both mental illness and indicators of social marginalization, not on all people receiving mental health care. Evidence that group-based interventions are superior to individually delivered interventions in other populations remains mixed[25,103–106]. The present findings should therefore not be generalized beyond the target population of the review without caution.

The quality of the evidence was generally strong. Most studies included in the meta-analysis were randomized trials, and many were preregistered. Most effect sizes were rated as having low or moderate risk of bias, and only one study was excluded from the synthesis because of critical risk of bias. Nonetheless, studies were typically small, reflecting the difficulty of recruiting this population. We mitigated some small-study concerns (e.g., the impact of chance bias[107]) by using baseline- or covariate-adjusted effect sizes for nearly all included effects. A more important limitation is that all included studies measured outcomes within one year of the intervention. The evidence, therefore, speaks to short-term effects only and cannot determine whether gains persist, fade, or grow over time.

Still, the findings have direct implications for practice and policy. Mental health systems in many high-income countries face rising demand while hospital-based care is increasingly replaced by community services. Group-based interventions are attractive in this context because they can increase treatment intensity and structure at a lower cost than individually delivered care. Our findings suggest that such interventions are not merely cheaper substitutes: on average, they improve both social reintegration and mental health. However, most control conditions were treatment as usual rather than the same intervention delivered individually. Thus, some benefits may reflect greater intensity, organization, or face-to-face contact rather than the group format alone. Policymakers should therefore view group-based interventions as a promising strategy for strengthening community care, while recognizing that the precise active ingredients remain uncertain.

On this background, future research in this field must address five priorities. First, studies should compare the same intervention delivered individually and in groups to isolate the added value of the group format. Second, a longer follow-up is needed to determine whether effects persist and which intervention features support durable recovery. Third, future work should examine the heterogeneity of participant experience, including whether some individuals benefit less or experience harm from group-based formats. Fourth, future research should identify the factors that explain variation in intervention effects. Fifth and finally, formal economic evaluations are also needed to quantify cost-effectiveness for adults facing both mental illness and social marginalization.

Materials and Methods

Protocol and preregistration

The protocol of this systematic review was peer-reviewed and pre-registered in Campbell Systematic Reviews[33]. We followed the protocol to the greatest extent possible. However, we made a small number of protocol deviations, primarily to incorporate newer methods with better statistical performance than those available when the protocol was submitted. These updates reflected recent methodological developments and were implemented to keep the analyses aligned with current best practice. All data and code are openly available, enabling replication and reanalysis using the originally proposed methods.

Two further clarifications were made. We did not add a ‘critical’ risk-of-bias category to RoB 2, as the tool guidance does not permit modification, and we now distinguish more clearly between primary social reintegration outcomes and secondary mental health outcomes than in the original protocol.

Eligibility criteria

Types of studies and settings

The review included RCTs, cluster RCTs, quasi-randomized trials, and non-randomized studies[108] with a well-defined control group. Non-randomized studies had to demonstrate baseline equivalence or adequately control for key confounders. Baseline equivalence was assessed using Cochrane RoB2 for randomized studies and ROBINS-I for non-randomized studies. Single-group pre–post studies and studies comparing two group-based interventions without a non-group control condition were excluded.

Eligible interventions had to be delivered in community or outpatient settings and aim to support participants’ social reintegration. We excluded interventions delivered during inpatient hospital care, but included outpatient group-based services following hospitalization or delivered within psychiatric or hospital settings.

Type of participants (P)

The population of this review consists of adults in OECD countries with at least one psychiatric diagnosis who are experiencing personal and social problems in addition to their mental health condition. We included participants with any type of psychiatric diagnosis, drawing on both studies in which patients self-reported their diagnosis and studies in which diagnoses were established by a mental health professional. Social and personal problems are defined broadly.

We excluded studies of interventions targeting youth under the age of 18. Psychiatric patients, without any co-morbid personal and social problems, who received outpatient treatment for their specific mental disorder with symptom reduction as the primary aim, were also not eligible for this review.

Type of interventions (I)

We applied a broad definition of group-based interventions. This included all interventions targeting adults who suffer from mental illness and related social and personal problems that received an intervention delivered in a group format. Specifically, this means that more than one participant should have received the intervention simultaneously, in the same physical setting, and from the same therapist, caseworker, mentor, etc. In addition, interventions were required to take place in a community or outpatient setting.

Type of control group (C)

We included three control types: individually delivered versions of the intervention, treatment as usual, and wait-list comparisons. Studies comparing two group-based interventions without a non-group control condition were excluded.

Type outcome measures (O)

We analyzed outcomes separately as primary social reintegration outcomes and secondary mental health outcomes. For social reintegration, the outcome included measures of social functioning, loneliness, hope, empowerment, self-efficacy, subjective well-being, quality of life, self-esteem, homelessness, employment, hospital admissions, and alcohol or substance use. Mental health outcomes included measures of general mental health, anxiety, depression, and psychosis symptoms. Find a full list of all included measures and scales in the SI Appendix.

We included all follow-up time points, but the available literature reported only post-test effects, defined per-protocol as outcomes measured within one year after the intervention ended.

Search methods for the identification of studies

The following databases were searched electronically: MEDLINE, EMBASE (OVID) 1974 – 2022, APA PsycINFO (EBSCO), CINAHL (EBSCO), Sociological Abstracts (ProQuest), Social Services Abstracts (ProQuest), SocINDEX (EBSCO), Academic Search Premier (EBSCO), International Bibliography of the Social Sciences (IBSS) (ProQuest), Science Citation Index (Web of Science Core Collection), Social Sciences Citation Index (Web of Science Core Collection), Cochrane Central Register of Controlled Trials (CENTRAL). The electronic database searches were completed in September 2022, and other resources were completed in July 2024. We searched to identify both published and unpublished literature. The searches were international in scope. In addition to electronic searches, we implemented a wide range of search methods and strategies to maximize coverage of relevant references. Find further details in the SI Appendix. The full search string can be found in the protocol[33].

Screening and data extraction

All title and abstract screening as well as full-text screening were conducted independently by at least two review authors, and all disagreements were resolved by either N.T.D and/or M.H.V. Exact screening performances can be found in Vembye et al.[109] Figure 2. Studies considered eligible by at least one team member or studies where there was insufficient information in the title and abstract to judge eligibility were retrieved in full text.

Data extraction was done in collaboration between N.T.D, J.S.A, J.K.J, and M.H.V, with minor support from two research assistants. All coding and data extraction were done independently by at least two reviewers. Before initiating the final extraction, our extraction scheme was piloted to ensure a standardized use thereof. Throughout the entire process, any extraction disagreements were resolved by N.T.D and/or M.H.V.

Effect sizes were primarily extracted and calculated by J.K.J and M.H.V, and quality checked by J.S.A in accordance with the prescribed procedures for ensuring reproducible research in statistics developed by Hofner et al.[110]. All effect size issues were resolved by M.H.V.

Risk of bias assessment

Risk of bias was assessed independently by at least two reviewers, with disagreements resolved by N.T.D or M.H.V. Assessments were conducted at the effect-size level to account for studies contributing multiple outcomes. We primarily assessed risk of bias for the treatment-on-the-treated effect, reflecting our focus on the effect of starting and adhering to the intended intervention.

Randomized and cluster-randomized trials were assessed using Cochrane's RoB 2[96], covering randomization, deviations from intended interventions, missing data, outcome measurement, and selective reporting. Non-randomized studies were assessed using Cochrane's ROBINS-I[97], with particular attention to confounding, participant selection, intervention classification, deviations from intended interventions, missing data, outcome measurement, and selective reporting.

For non-randomized studies, we assessed whether authors established baseline equivalence or adequately controlled for key confounders, including pretest scores, age, gender, psychiatric diagnosis, and comorbid social or personal problems. Across tools, studies with four non-low domain judgments were moved to the next overall risk category. Following the tool guideline, studies judged to be at critical risk of bias were excluded from the meta-analysis.

Effect size calculation

Treatment effects were estimated by comparing group-based interventions with eligible control conditions. We used standardized mean differences, calculated as Hedges' *g*,[111–115] and coded all effects so that positive values favored group-based interventions. Odds ratios were calculated for one study only and were not analyzed further.

For all but one study[83], effect sizes were baseline- or covariate-adjusted to improve precision and internal validity.[116] Where studies reported results for non-prespecified subgroups of the review, these were aggregated to reduce artificial within-study variation.[117] When more than five studies reported effects using the same scale, we computed the population-based standard deviation obtained from the control groups[118] to increase the generalizability.

Because group-based interventions create dependence among participants treated in the same group, we adjusted effect sizes and variances for partial clustering in the intervention arm.[119] When intraclass correlations were not reported, we imputed an ICC of 0.10 in the main analyses and tested alternative values in sensitivity analyses. All formulas behind this calculation can be found at https://mikkelvembye.github.io/VIVECampbell/reference/vgt_smd_1armcluster.html. Although cluster-bias correction is recommended for group-delivered treatments[120], both adjusted and unadjusted standard errors may be biased[121]. We therefore tested robustness using a non-cluster-adjusted version of the *g* estimator.

All effect size calculations for each study can be found in the Online Appendix A, and further details about the statistical procedure can be found in the SI Appendix.

Statistical synthesis of effect sizes

Overall average effects

Because studies often contributed multiple dependent effect sizes, we estimated overall effects using the partially empirical correlated-hierarchical effects model with robust variance estimation (PECHE-RVE).[122] This model accounts for effect sizes nested within studies, estimates heterogeneity at both study and effect-size levels, and adjusts for correlated sampling errors. Within-study correlations were estimated where possible (hence the term "partially empirical") and otherwise imputed as ρ = 0.8, as specified in the protocol. Robust variance estimation was used to reduce sensitivity to model misspecification.[123,124]

We assessed heterogeneity using the between-study (τ) and within-study (ω) standard deviations, as well as the total true-effect variation $\left(\sqrt{\tau^2 + \omega^2}\right)$, I^2 and 67% prediction intervals for the main effects. See the SI Appendix for further model heterogeneity details as well as how we defined the unit of analysis.

Sensitivity analysis

We tested robustness by varying the assumed within-study effect-size correlation, effect-size metric, clustering adjustment, and intraclass correlation, switching TOT with ITT measures for the three studies reporting both, and by excluding high-risk-of-bias effects and non-randomized studies. We did not conduct outlier sensitivity analyses because no effect sizes met our outlier definition.[125,126]

Publication bias analysis

We assessed publication bias and small-study effects using multiple complementary methods, as no single test is optimal across all levels of selection. Analyses were conducted separately for social reintegration and mental health outcomes and included the hybrid extended meta-analysis (HYEMA)[102], worst-case sensitivity analyses[127], puniform*[128], PECHE-ISCW[99], PET/PEESE-ISCW[99], and bootstrapped selection models[129]. Contour-enhanced funnel plots[130] were used for visualization. All models accounted for dependent effect sizes and preregistration status.

We examined whether publication-bias adjustments altered subgroup effects by preregistration status or outcome type. Modified standard errors were used where appropriate to avoid artificial associations between standardized mean differences and their sampling variances[131]. Find further model details in the SI Appendix.

Moderator analysis

We examined heterogeneity using subgroup and moderator analyses for theoretical, methodological, and bias-related factors. Categorical moderators were analyzed with the partially empirical subgroup correlated-effects plus (PESCE+[-RVE]) models with robust variance estimation and cluster wild bootstrapping[132], while continuous moderators were analyzed with PECHE-RVE models. Models were estimated with and without covariate adjustment, controlling for key outcome, sample, intervention, and design characteristics where multicollinearity permitted. We also used the partial empirical correlated multivariate effects (PECMVE) model to test whether effects on social reintegration and mental health covaried within studies. See SI Appendix for the full list of all included moderators and further model details.

Used software

The review used *EPPI-Reviewer* for screening and study management. Microsoft Excel was used for extracting covariates and background data, and R (4.5.1, 4.5.2, and 4.6.0) and RStudio for effect-size calculation, meta-analysis, and visualization. Key R packages included *tidyverse*[133] for data management, *estmeansd*[134] for effect-size inputs from medians and quantiles, *metafor*[135], *clubSandwich*[136], and *wildmeta*[137] for meta-analysis and robust inference, *ggplot2*[138] for visualization, and our own *VIVECampbell*[139] package for cluster-bias adjustments. Publication-bias analyses used *boot*[140], *metaselection*[141], and *puniform*[142].

Data availability

All data and materials behind this study can be found at https://osf.io/s2j9a. This includes search string examples, all data extractions, risk of bias assessments, effect size calculations, and meta-analysis code and data, etc. A full variable description is available in the second sheet of the gb_dat.xlsx data.

Code availability

All codes can be found in the following online material stored at https://osf.io/s2j9a:

- Appendix A: Effect size calculation
- Appendix B: Data manipulation, RoB visualization, and PRIMED workflow
- Appendix C: Main results code
- Appendix D: Sensitivity analysis code
- Appendix E: Publication bias code

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Figures and Tables

Figure 1. PRISMA flowchart of the screening process

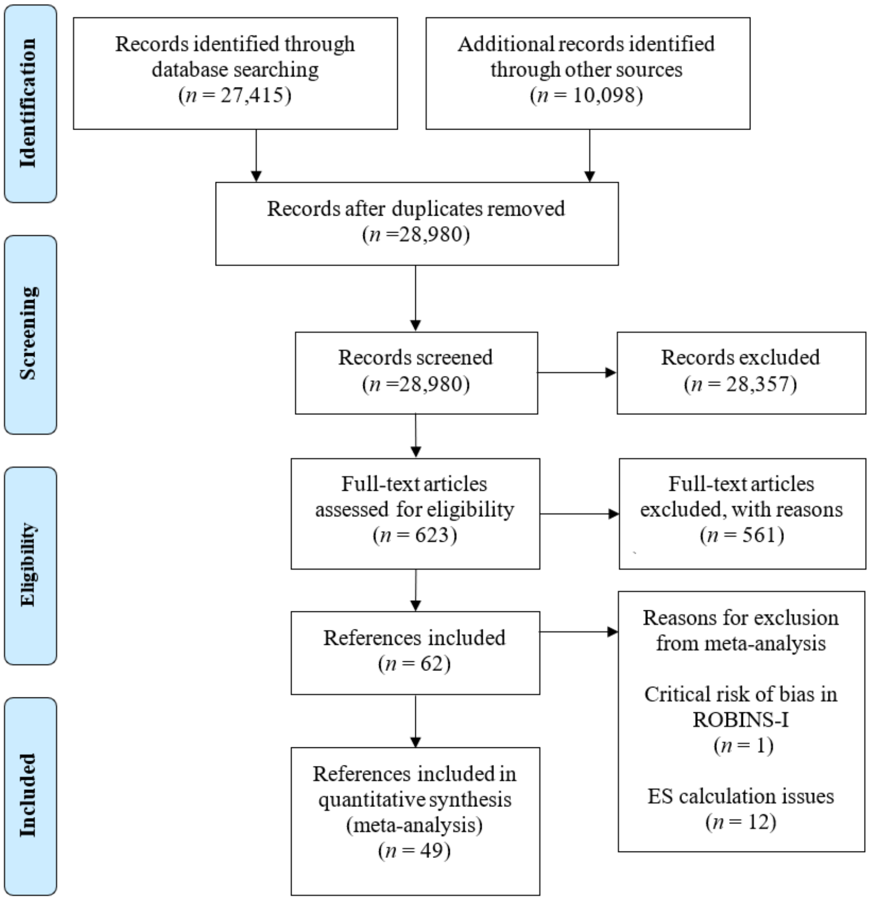


Figure 2. Number of studies included in meta-analysis by year and registration status

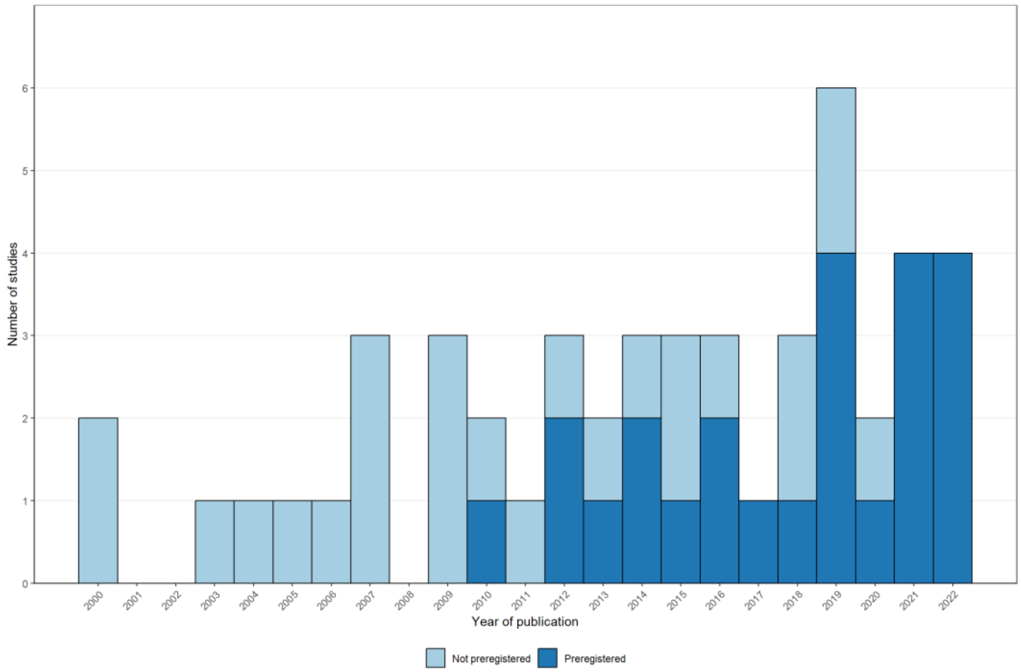


Figure 3. Effect size forest plot across dependent effect sizes by type of outcome

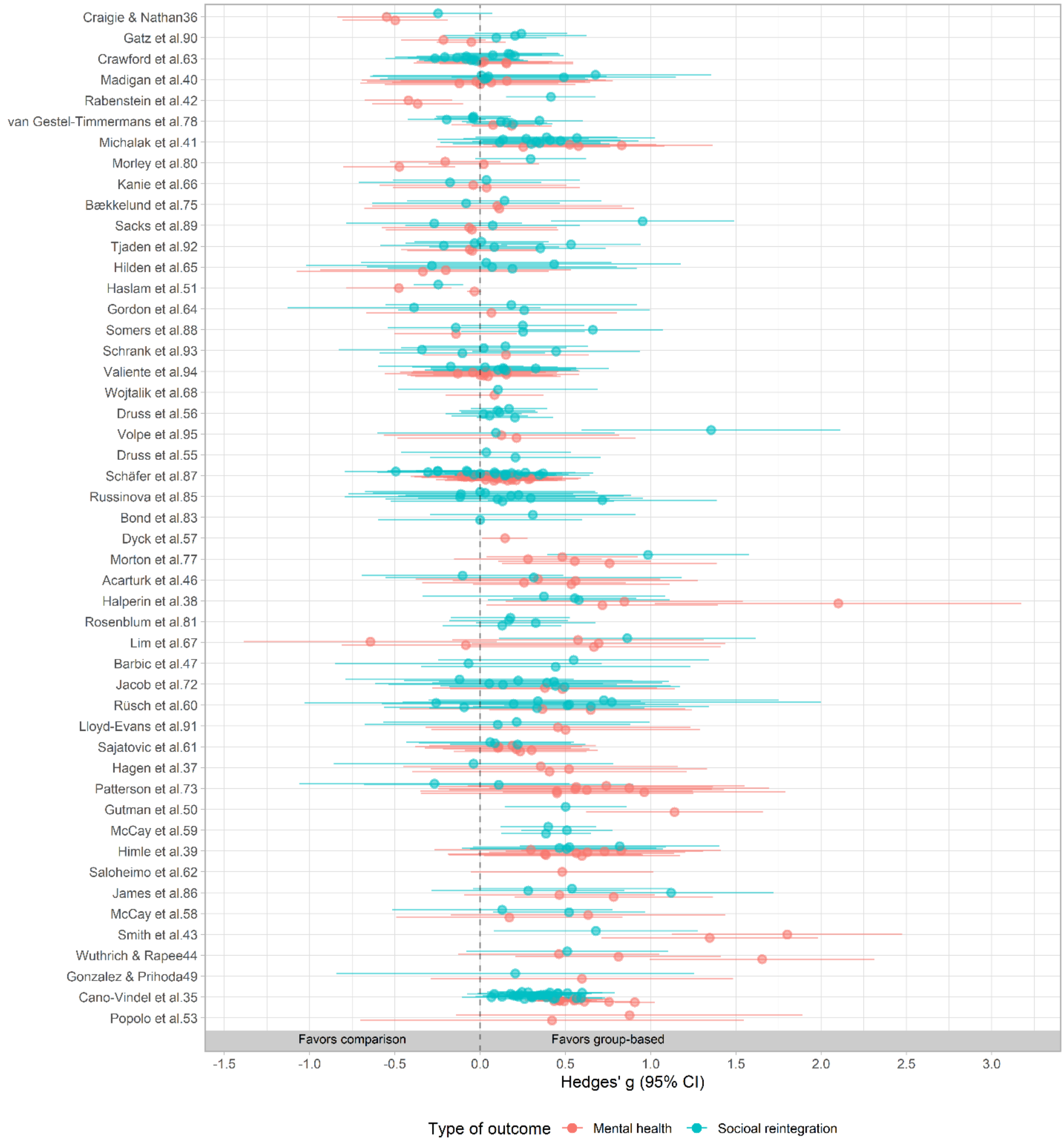
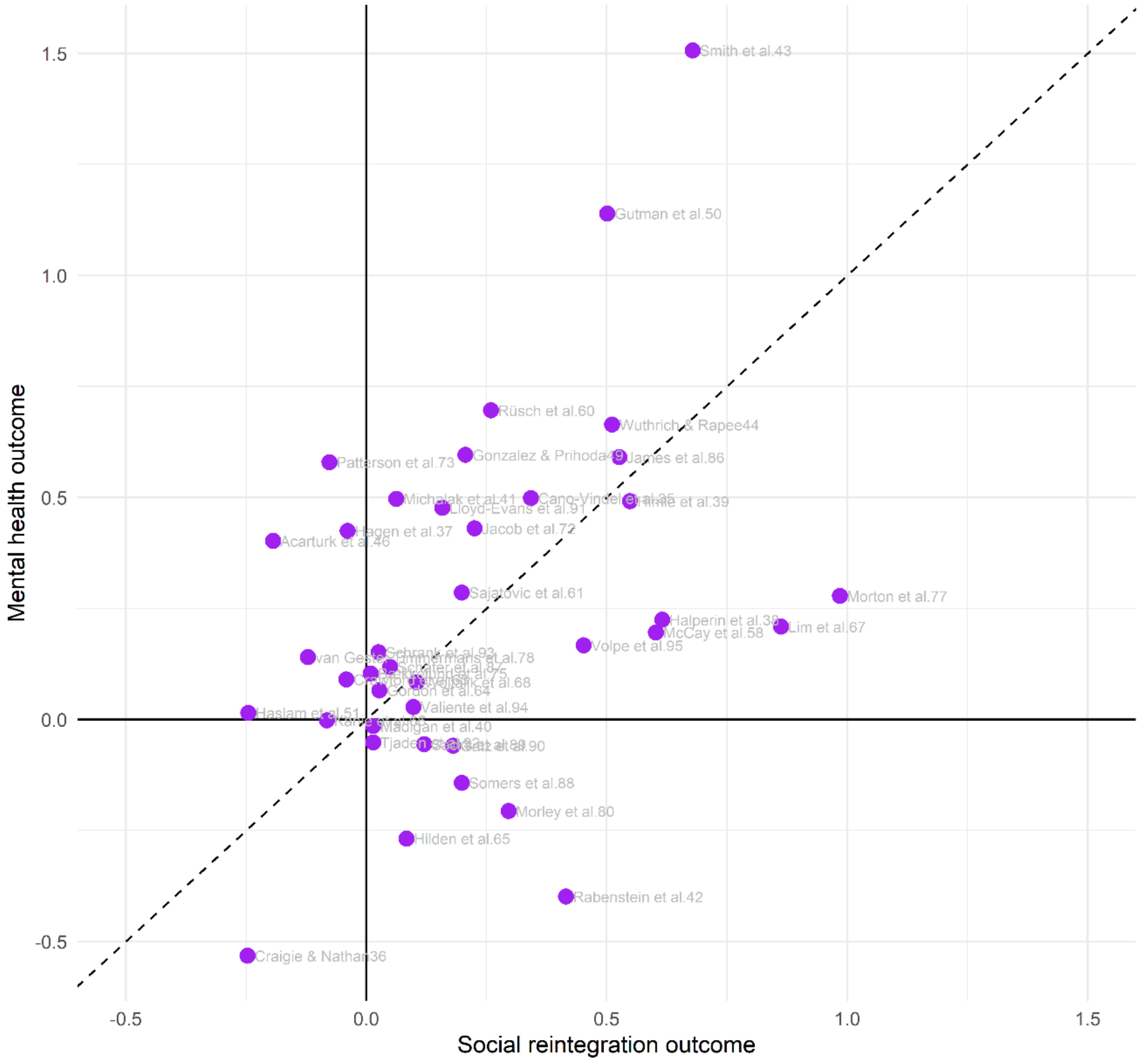
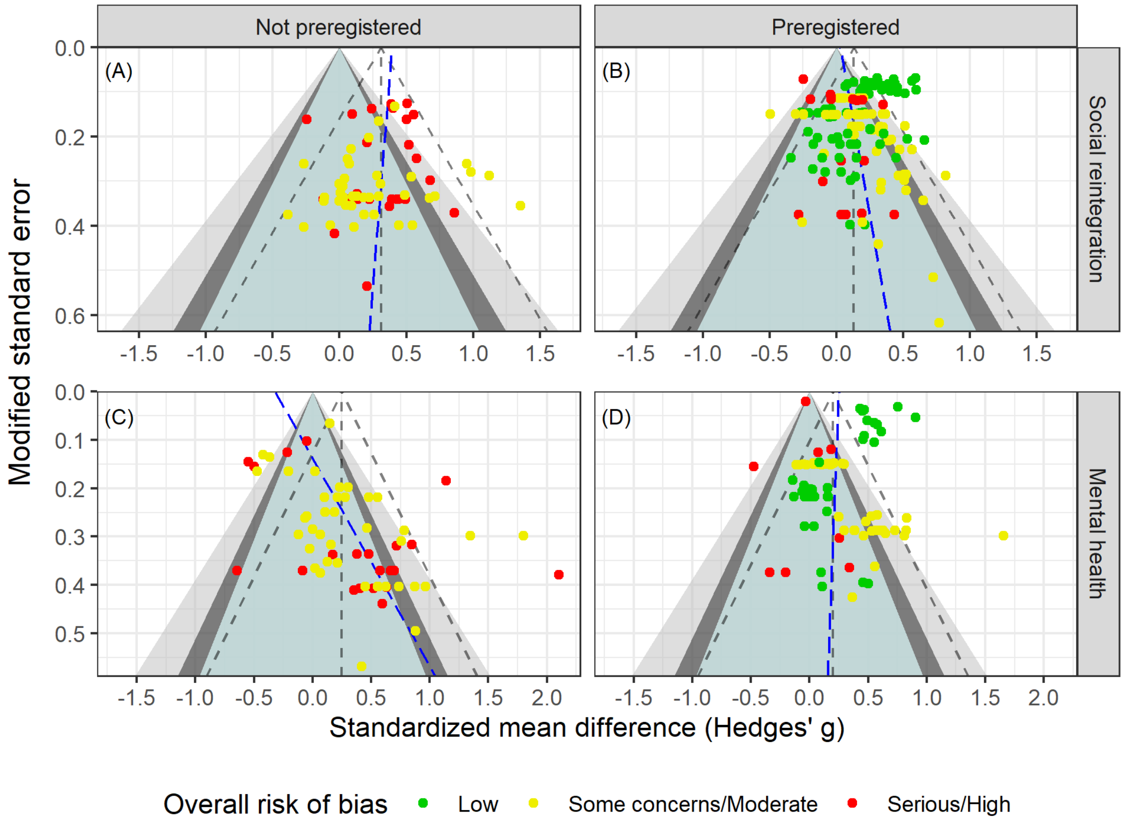


Figure 4. Scatter plot illustrating the aggregated effects of social reintegration and mental health outcomes



Note. This plot is based on the 39 studies that both report social integration and mental health outcomes. The dashed line indicates equal effects on social reintegration and mental health.

Figure 5. Funnel plot across registration status and type of outcome



Note. Following van Aert[102], slopes at the effect size level were obtained from the PECHE-RVE[99] model. Effects in the green region have p-values above 0.1, effects in the dark gray area have p-values between 0.05 and 0.1, effects in the light gray area contain p-values ranging from 0.01 to 0.05, and finally, effects in the white area have p-values less than 0.01.

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Table 1. Included group-based interventions.

Intervention category	Studies	In meta-analysis
Group-based cognitive behavioral therapy[34–45]	12	10
Group psychoeducation and social skills training[46–54]	9	6
Illness management[55–62]	8	8
Social cognition and interaction training[63–68]	6	6
Cognitive-behavioral social skills training[41,69–73]	6	3
Group psychoeducation[74–78]	5	3
Illness and addiction management[79–82]	4	2
Vocational training[83–85]	3	2
Addiction management[86,87]	2	2
Residential treatment[88,89]	2	2
Seeking Safety[87,90]	2	2
Social network training[91,92]	2	2
Positive psychology group intervention[93,94]	2	2
Art therapy[63]	1	1
Reading group intervention[95]	1	1

Note. The numbers might sum to more than 62 (i.e., 65) and 49 (i.e., 52) studies included meta-analysis, as three studies contributed with two treatments each.

Table 2. Descriptive percentages for the included studies with reintegration outcomes

Characteristics	Studies (j)	Effects (m)	% Studies	% Effects
Context				
Australia	9	15	19.6	7.3
Austria	1	1	2.2	0.5
Canada	4	12	8.7	5.9
Korea	1	1	2.2	0.5
Finland	1	5	2.2	2.4
Germany	4	46	8.7	22.4
Ireland	1	6	2.2	2.9
Italy	1	2	2.2	1.0
Japan	1	2	2.2	1.0
Netherlands	2	14	4.3	6.8
Norway	2	3	4.3	1.5
Spain	2	35	4.3	17.1
Turkey	1	2	2.2	1.0
UK	3	19	6.5	9.3
US	13	42	28.3	20.5
Outcome measure				
Alcohol and drug abuse/misuse	8	32	17.4	15.6
Employment	1	2	2.2	1.0
Hope, empowerment & self-efficacy	12	32	26.1	15.6
Loneliness	4	5	8.7	2.4
Physical health	2	3	4.3	1.5
Psychiatric hospitalization	1	1	2.2	0.5
Self-esteem	5	14	10.9	6.8
Social functioning (degree of impairment)	17	48	37.0	23.4
Well-being and quality of life	25	68	54.3	33.2
Diagnosis in the sample				
Schizophrenia	9	33	19.6	16.1
Other	37	172	80.4	83.9
Intervention type				
CBT	12	55	26.1	26.8
Other	35	150	76.1	73.2
Preregistration status				

Not preregistered	24	65	52.2	31.7
Preregistered	22	140	47.8	68.3
Outlet				
Journal article	46	205	100.0	100.0
Research design				
QES	8	18	17.4	8.8
RCT	38	187	82.6	91.2
Test type				
Clinician-rated measure	12	36	26.1	17.6
Raw events	1	1	2.2	0.5
Self-reported	39	168	84.8	82.0
Analysis strategy				
ITT	22	108	47.8	52.7
TOT	24	97	52.2	47.3
Control group				
Individual treatment	3	4	6.5	2.0
TAU	30	154	65.2	75.1
TAU with Waiting-list	9	34	19.6	16.6
Waiting-list only	4	13	8.7	6.3
Overall risk of bias				
Low	10	69	21.7	33.7
Some concerns/Moderate	21	94	45.7	45.9
Serious/High	16	42	34.8	20.5
Timing of measurement				
Short-term measure (< 52 weeks)	46	205	100.0	100.0
Effect size metric				
Posttest only ES	1	2	2.2	1.0
Pre-posttest/covariate adj. ES	45	203	97.8	99.0
Handles multilevel structure				
No	43	186	93.5	90.7
Yes	3	19	6.5	9.3

Note: ES = Effect size, ITT = Intention-To-Treat; TOT; Treatment-On-the-Treated; QES = Quasi-Experimental Design (non- randomized); RCT = Randomized Controlled Trial; TAU = Treatment as usual; CBT = cognitive behavioral therapy

Supporting Information for ”A Meta-Analysis of the Effects of Group-Based Interventions on Social Reintegration of Marginalized Adults with Mental Illness”

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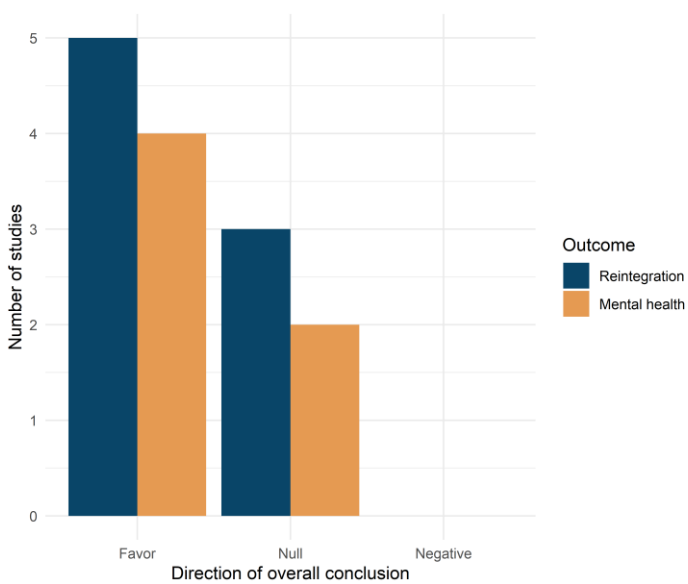
Potential biases in the review process

We performed a comprehensive electronic database search, combined with grey literature searching, backwards citation tracking, and hand searching of key journals. All citations were screened by two independent screeners from the review team, and one review author (NTD) assessed all included studies against the inclusion criteria. We believe that nearly all the publicly available studies on the effects of group-based interventions for marginalized adults suffering from both mental illness and social problems within the OECD countries were identified during the review process. However, 35 references were not obtained in full text. This can potentially be a source of bias in the review. Similarly, it can have induced bias that we were not able to include all studies, as we could not calculate the effect size for 12 studies. However, as illustrated in Figure S1, we did not find any indication that the 12 excluded studies reached a different conclusion than our review.

Figure S1. Bar plot illustrating directions of overall conclusions from the 12 studies without effect sizes



Further descriptive statistics

Table S1 describes the included participants by primary type of mental disorder and primary indicator of marginalization. Table S2 presents descriptive averages for studies that reported social reintegration outcomes. Tables S2 and S3 present descriptive statistics for studies reporting mental health outcomes.

Table S1. Characteristics of participants

Characteristics of the participants	Studies	In meta-analysis
Mental health disorder		
Schizophrenia or other primary psychotic disorders	36	25
Depressive disorders	25	23
Bipolar disorders	20	15
Anxiety disorders	15	14
Personality disorders[a]	10	9
Post-traumatic stress disorder	9	9
Other mental health disorders[b]	9	9
Borderline personality disorder	8	4
Obsessive-compulsive disorder	3	3
Not specified	5	3
Social marginalization		
Substance abuse[c]	13	11
Other social problems[e]	6	6
Chronic medical conditions[d]	6	4
Unemployment	5	2
Trauma experience	3	3
Homelessness	2	2
Loneliness	2	2
Psychiatric comorbidity	1	1
Criminal justice involvement	1	0
Not clearly specified	28	21

Note. Participants with different mental health disorders and indicators of social marginalization could appear within the same study. Therefore, the categories are not mutually exclusive, and the columns sum to more than 62 and 49 studies, respectively.

[a] If specified, primarily DSM cluster C personality disorders.

[b] Including dissociative identity disorder, autism spectrum disorder, eating disorders, other psychotic disorders, suicidal ideation, and elevated levels of psychological stress

[c] Including both alcohol and drugs

[d] Including both psychiatric and somatic chronic conditions

[e] Including living in residential group homes, being older than 60 years old, and self-reported social problems such as having experienced disruptive periods, elevated internalized stigma, and experiencing social or cognitive disability

Table S2. Descriptive means of included studies reporting social reintegration outcomes

Characteristics	*Studies (j)*	*Effects (k)*	$Median_j$	$Mean_j$	SD_j	*Range*
Sample characteristics						
Raw treatment sample size	46	205	29.00	58.00	60.80	7-315
Raw control sample size	46	205	30.00	54.00	60.20	9-316
Total sample size	46	205	60.00	118.00	127.20	16-631
Effective treatment sample size	46	205	19.00	34.00	35.00	4-175
Effective control sample size	46	205	19.00	32.00	34.60	5-176
Total effective sample size	46	205	38.00	70.00	73.30	10-351
Average age in studies	46	205	40.80	40.70	8.90	25-67
% Males	45	202	46.30	46.60	23.40	0-81
Intervention characteristics						
Average group size	27	133	8.00	7.90	3.20	4-22
Total number of sessions	45	201	12.00	21.00	21.90	3-104
Sessions per week	45	201	1.00	1.30	1.50	0-10.5
Duration (in weeks)	46	205	12.00	18.30	16.90	4-78
Measurement timing						
After end of intervention (in weeks)	46	205	0.50	9.80	15.90	0-52
From baseline (in weeks)	46	205	22.00	28.10	26.10	6-130
Methodological features						
Pre-posttest correlation (empirically computed)[a]	9	24	0.76	0.62	0.22	0.187-0.915

Note.
[a] Computed at the effect size level

Table S3. Descriptive percentages for the included studies for mental health outcomes

Characteristics	Studies (j)	Effects (k)	% Studies	% Effects
Context				
Australia	9	22	21.4	15.3
Austria	1	2	2.4	1.4
Canada	2	3	4.8	2.1
Korea	1	5	2.4	3.5
Finland	2	3	4.8	2.1
Germany	4	26	9.5	18.1
Ireland	1	6	2.4	4.2
Italy	2	4	4.8	2.8
Japan	1	2	2.4	1.4
Netherlands	2	4	4.8	2.8
Norway	2	5	4.8	3.5
Spain	2	21	4.8	14.6
Turkey	1	4	2.4	2.8
UK	3	7	7.1	4.9
US	9	30	21.4	20.8
Outcome measure				
Anxiety	9	14	21.4	9.7
Depression	20	37	47.6	25.7
General mental health	29	72	69.0	50.0
Symptoms of psychosis	7	21	16.7	14.6
Diagnosis in sample				
Schizophrenia	7	14	16.7	9.7
Other	35	130	83.3	90.3
Intervention type				
CBT	12	49	28.6	34.0
Other	31	95	73.8	66.0
Preregistration status				
Not preregistered	22	62	52.4	43.1
Preregistered	20	82	47.6	56.9
Outlet				
Journal article	42	144	100.0	100.0
Research design				
QES	8	17	19.0	11.8
RCT	34	127	81.0	88.2
Test type				

Characteristics	Studies (j)	Effects (k)	% Studies	% Effects
Clinician-rated measure	14	41	33.3	28.5
Self-reported	33	103	78.6	71.5
Analysis strategy				
ITT	18	69	42.9	47.9
TOT	24	75	57.1	52.1
Control group				
Individual treatment	2	3	4.8	2.1
TAU	28	108	66.7	75.0
TAU with Waiting-list	9	26	21.4	18.1
Waiting-list only	3	7	7.1	4.9
Overall risk of bias				
Low	10	36	23.8	25.0
Some concerns/Moderate	20	79	47.6	54.9
Serious/High	13	29	31.0	20.1
Timing of measurement				
Short-term measure (< 52 weeks)	42	144	100.0	100.0
Effect size metric				
Pre-posttest adj. ES	42	144	100.0	100.0
Handles multilevel structure				
No	40	140	95.2	97.2
Yes	2	4	4.8	2.8

Note: ES = Effect size, ITT = Intention-To-Treat; TOT; Treatment-On-the-Treated; QES = Quasi-Experimental Design (non-randomized); RCT = Randomized Controlled Trial; TAU = Treatment as usual; CBT = cognitive behavioral therapy.

Table S4. Descriptive means of included studies reporting mental health outcomes

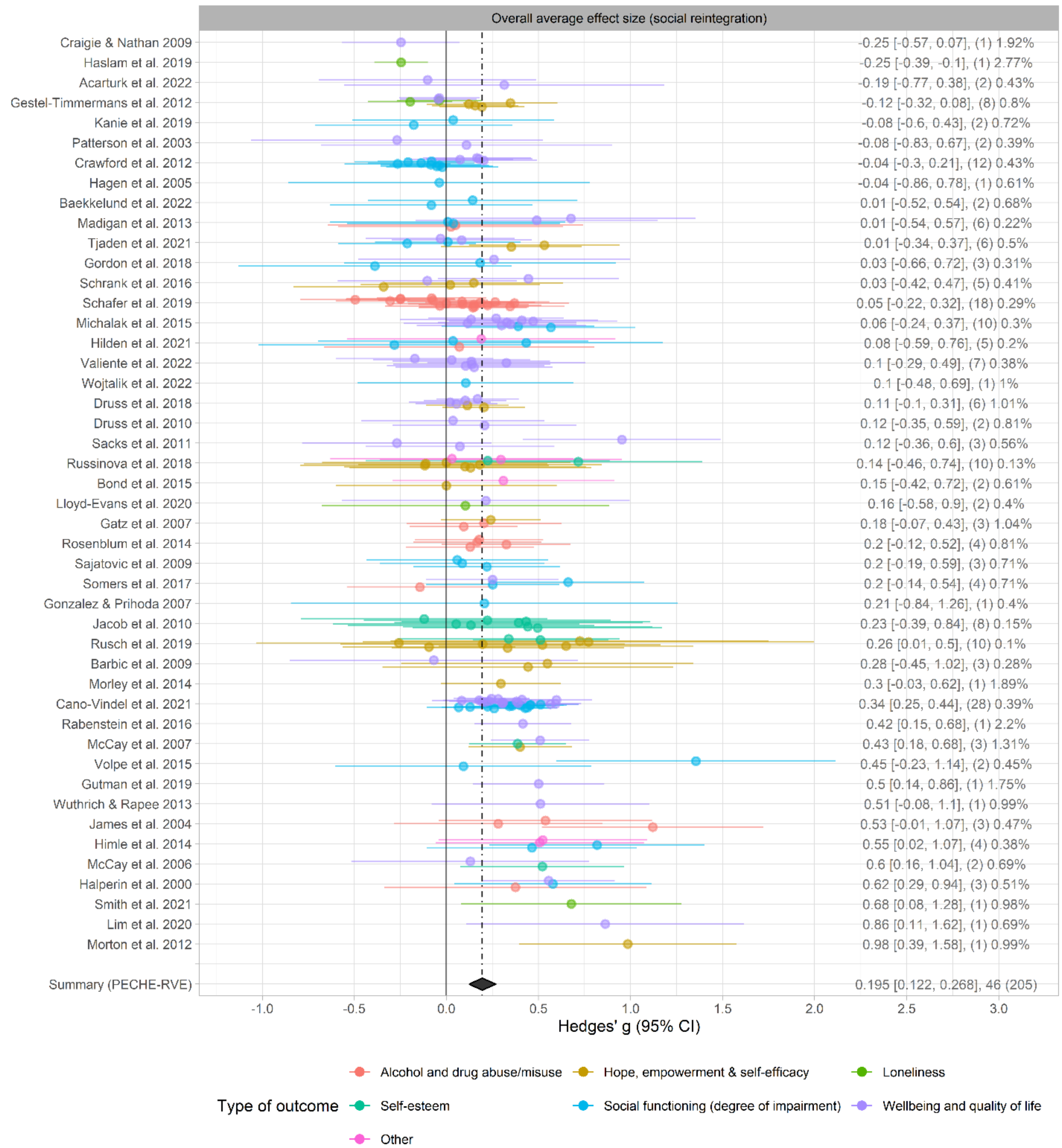
Characteristics	*Studies (j)*	*Effects (k)*	$Median_j$	$Mean_j$	SD_j	*Range*
Sample characteristics						
Raw treatment sample size	42	144	28.00	54.00	59.40	7-315
Raw control sample size	42	144	29.00	51.00	58.30	9-316
Total sample size	42	144	60.00	111.00	125.00	16-631
Effective treatment sample size	42	144	20.00	33.00	33.90	4-175
Effective control sample size	42	144	19.00	30.00	33.20	5-176
Total effective sample size	42	144	38.00	67.00	71.50	10-351
Average age in studies	42	144	39.70	39.60	9.30	22-67
% Males	41	143	46.30	45.30	23.50	0-81
Intervention characteristics						
Average group size	26	92	7.00	7.10	1.60	4-10
Total number of sessions	41	143	13.00	21.60	22.50	3-104
Sessions per week	41	143	1.00	1.30	1.50	0-10.5
Duration (in weeks)	42	144	12.00	19.30	17.50	4-78
Measurement timing						
After end of intervention (in weeks)	42	144	1.00	10.70	16.60	0-52
From baseline (in weeks)	42	144	22.00	30.10	27.40	6-130
Methodological features						
Pre-posttest correlation (empirically computed)[a]	9	18	0.48	0.51	0.17	0.12-0.76

Note.
[a] Computed at the effect size level

Detailed forest plots

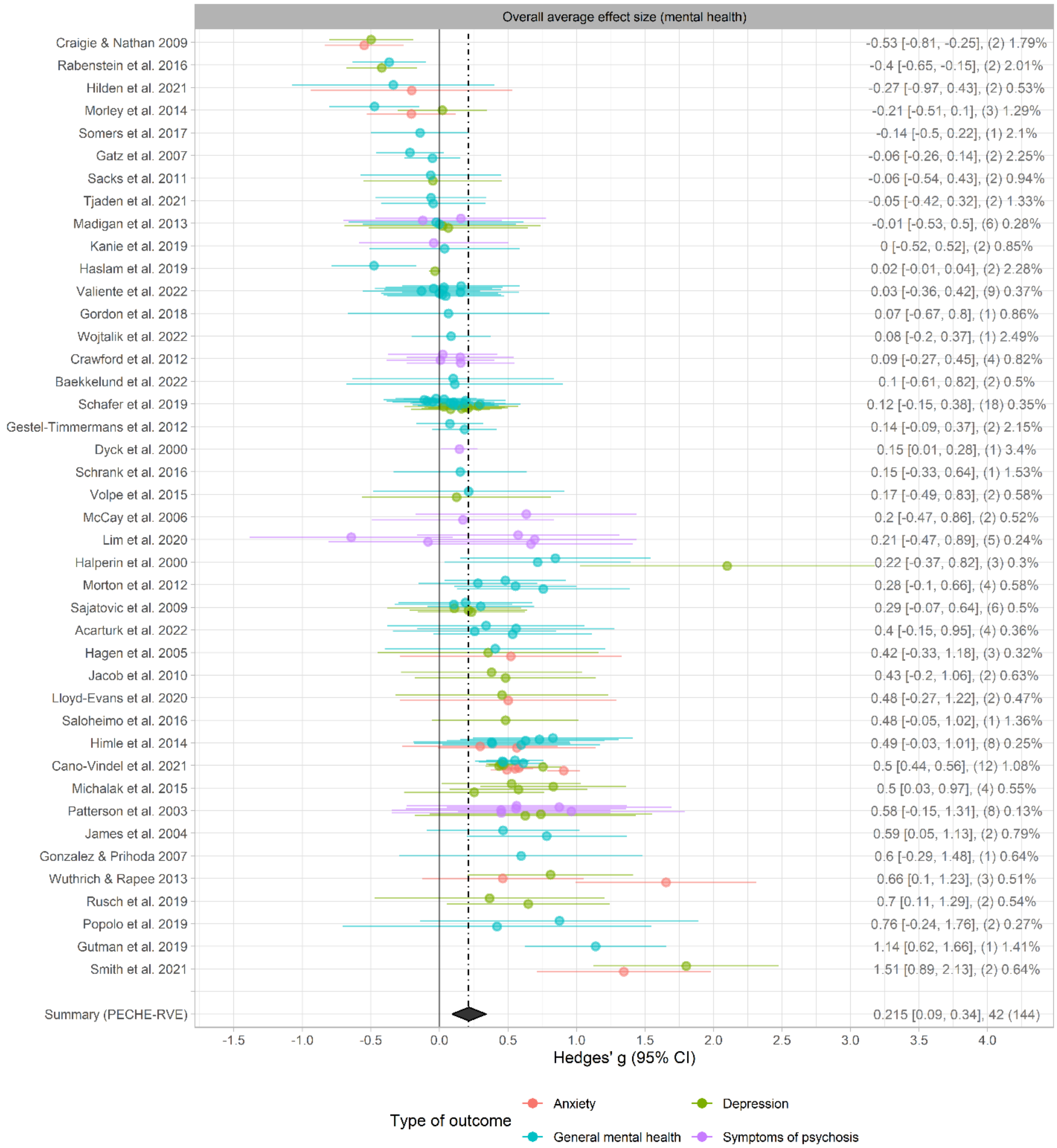
Figure S2 and S3 present more outcome-specific forest plot, including specific weight attributed to each effect size within the given study and the overall average effect size.

Figure S2. Effect size forest plot with dependent effect sizes for social reintegration outcomes



Note: The number of effect size estimates is in parentheses. The percentages indicate the weight given to each effect size within the given study. Studies are ordered by the study mean effect size obtained from fitting the within-study effect sizes to a univariate meta-analysis model, as suggested by Fernández-Castilla, Declercq, et al.[1] The dashed line indicates the overall average effect size (g = .195), and the diamond indicates the 95% confidence interval from the fitted PECHE-RVE model.

Figure S3. Effect size forest plot with dependent effect sizes for mental health outcomes

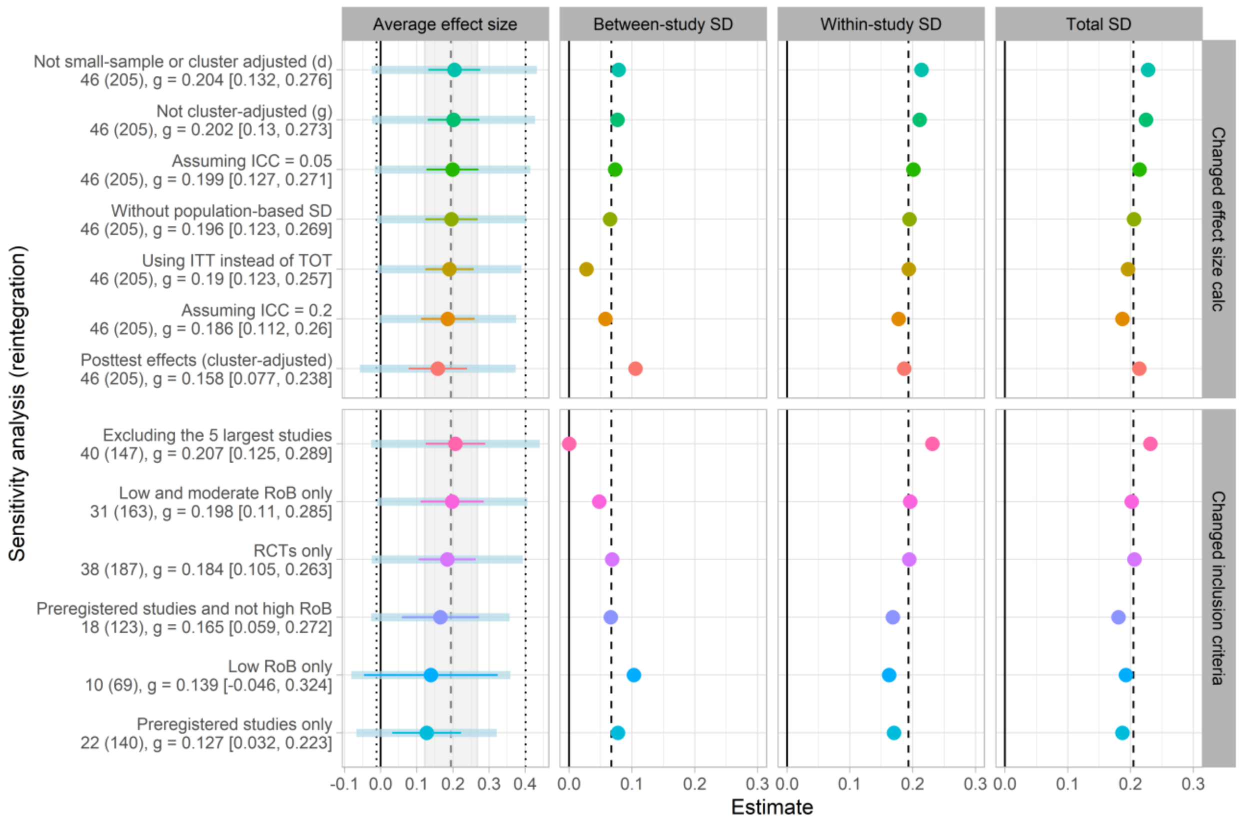


Note: The number of effect size estimates is in parentheses. The percentages indicate the weight given to each effect size within the given study. Studies are ordered by the study mean effect size obtained from fitting the within-study effect sizes to a univariate meta-analysis model, as suggested by Fernández-Castilla, Declercq, et al.[1] The dashed line indicates the overall average effect size (g = .215), and the diamond indicates the 95% confidence interval from the fitted PECHE-RVE.

Sensitivity analysis visualized

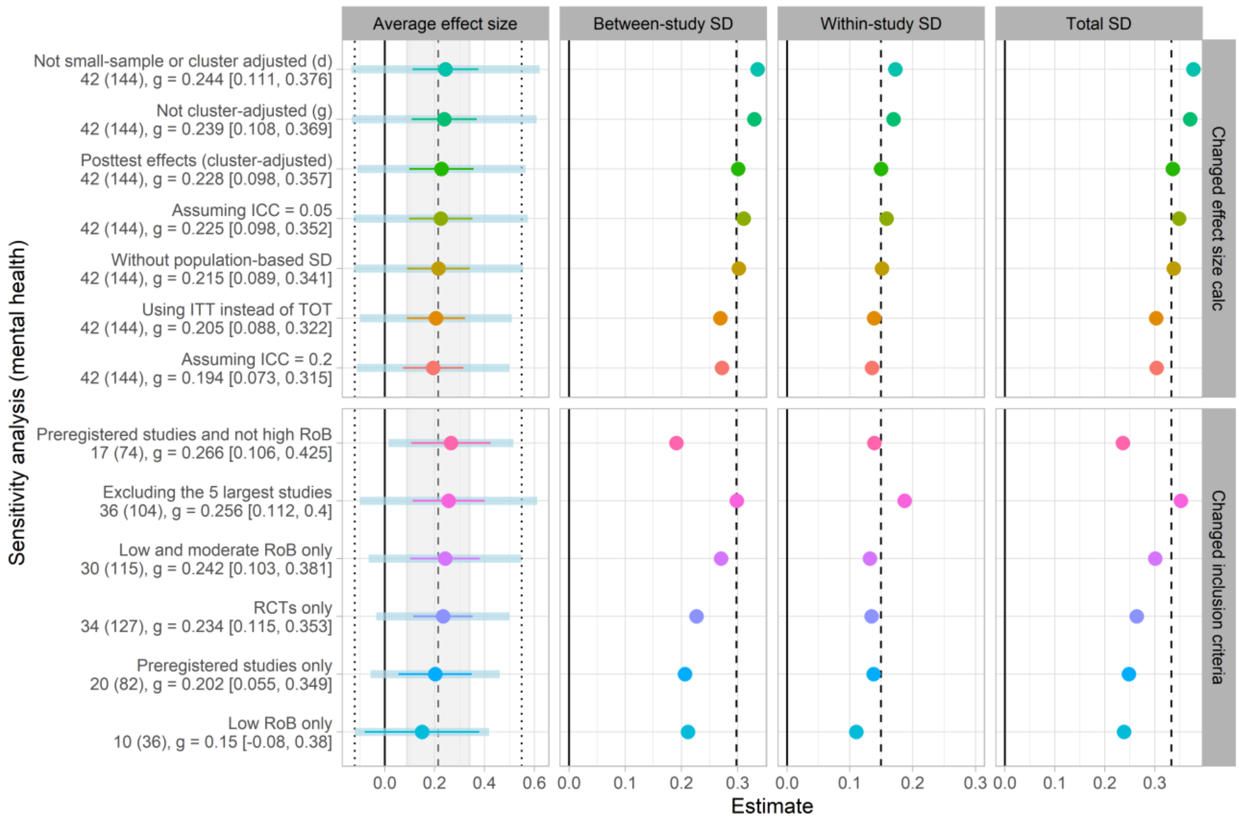
Figures S4-S5 present sensitivity analyses while changing effect size calculation assumptions and inclusion criteria for social reintegration and mental health outcomes, respectively.

Figure S4. Sensitivity analyses changing effect size calculation assumptions and inclusion criteria for social reintegration outcomes



Note: Dashed lines indicate the estimated values from the main analysis. The gray shading in the first column represents the width of the confidence interval from the main analysis, while the point lines indicate the re-estimated confidence intervals. Dotted lines indicate the 67% prediction interval estimated from the main analysis, and the blue bands in the first column represent the re-estimated 67% prediction intervals.

Figure S5. Sensitivity analyses changing effect size calculation and inclusion criteria for mental health outcomes



Note: Dashed lines indicate the estimated values from the main analysis. The gray shading in the first column represents the width of the confidence interval from the main analysis, while the point lines indicate the reestimated confidence intervals. Dotted lines indicate the 67% prediction interval estimated from the main analysis, and the blue bands in the first column represent the re-estimated 67% prediction intervals.

Publication bias table for the overall average effects

Table S5. Overall average effects across different publication bias-adjusted models

Analysis	Studies	Effects	Est [95% CI]	p-val	τ	ω	SD √(τ²+ω²)	λ_1	λ_2
Social reintegration									
Bootstrap HYEMA	46	205	0.187 [0.089, 0.195]	-	0.176	-	0.176	-	-
puniform*	46	205	0.245 [0.185, 0.312]	0.000	0.203	-	0.203	-	-
Worst-case meta-analysis	40	157	0.078 [0.025, 0.130]	0.006	0.000	0.149	0.149	-	-
PET/PEESE[a]	46	205	0.246 [-1.403, 1.895]	0.513	0.073	0.194	0.207	-	-
Bootstrap 3PSM	46	205	0.232 [0.135, 0.324]	0.000	0.186	-	0.186	1.478	-
Bootstrap 3PSM (non-preregistered)[b]	46	205	0.189 [0.082, 0.278]	0.000	0.176	-	0.176	0.524	-
Bootstrap 4PSM	46	205	0.189 [0.025, 0.319]	0.003	0.200	-	0.200	1.304	0.854
Bootstrap 4PSM (non-preregistered)[b]	46	205	0.176 [0.062, 0.273]	0.002	0.181	-	0.181	0.625	0.255
Mental health									
Bootstrap HYEMA	42	144	0.290 [0.131, 0.436]	-	0.325	-	0.325	-	-
puniform*	42	144	0.598 [0.454, 0.763]	0.000	0.421	-	0.421	-	-
Worst-case meta-analysis	37	108	0.069 [-0.031, 0.170]	0.168	0.169	0.100	0.196	-	-
PET/PEESE[a]	42	144	-0.063 [-1.489, 1.362]	0.880	0.315	0.172	0.359	-	-
Bootstrap 3PSM	42	144	0.426 [0.230, 0.737]	0.000	0.359	-	0.359	2.927	-
Bootstrap 3PSM (non-preregistered)[b]	42	144	0.289 [0.122, 0.456]	0.000	0.330	-	0.330	1.527	-
Bootstrap 4PSM	42	144	0.324 [0.026, 0.624]	0.001	0.393	-	0.393	2.565	1.296
Bootstrap 4PSM (non-preregistered)[b]	42	144	0.273 [0.102, 0.447]	0.001	0.338	-	0.338	1.677	1.108

Note. Confidence intervals for bootstrap models are percentile confidence intervals. HYEMA = Hybrid Extended Meta-Analysis. 3PSM = three-parameter selection model, 4PSM = four-parameter selection model, CHE = correlated-hierarchical effects, PET = precision effect test, PEESE = precision-effect estimator with standard error, τ = between-study variance, ω = within-study variance, λ = selection ratio parameter.

[a] Included control for preregistration.

[b] This cluster selection model only accounted for selection in non-preregistered studies.

Moderator analyses across social reintegration and mental health

In this section, we present all moderator analyses conducted for the review. The results can be found in Tables S6-S11, below.

Table S6. Substantive categorical moderators meta-analysis of social reintegration outcomes

Subgroup-analyses			Unadjusted effects				Covariate-adjusted effects[a]			
Moderator	Studies	Effects	Est. [95% CI]	p-val	Satt. df	SD $\sqrt{\tau^2 + \omega^2}$	Est. [95% CI]	p-val	Satt. df	SD $\sqrt{\tau^2 + \omega^2}$
Outcome type	46	205								
Alcohol/drug use	8	32	0.12 [-0.08, 0.31]	0.161	**3.6**	0.21	0.22 [0.06, 0.38]	**0.016**	6.7	0.21
Hope	12	32	0.23 [0.11, 0.34]	**0.003**	7.5	0.16	0.24 [0.09, 0.38]	**0.004**	10.2	0.16
Loneliness	4	5	0.02 [-0.63, 0.66]	0.938	**2.6**	0.31	-0.03 [-0.25, 0.2]	0.732	**3.2**	0.08
Self-esteem	5	14	0.42 [0.31, 0.54]	**0.001**	**3.4**	0.15	0.41 [0.24, 0.59]	**0.003**	4.2	0.15
Social functioning	17	48	0.18 [0.02, 0.34]	**0.027**	10.8	0.21	0.18 [0.02, 0.34]	**0.031**	11.4	0.19
Wellbeing & QoL	25	68	0.22 [0.12, 0.32]	**0.000**	13.0	0.18	0.2 [0.1, 0.29]	**0.000**	14.6	0.16
Other	4	6	0.32 [0.02, 0.62]	*0.044*	**2.8**	0.00	0.34 [-0.11, 0.79]	0.096	**3.2**	0.00
Wald test (CWB)				0.267				0.218		
Schizophrenia in sample	46	205								
Yes	9	33	0.19 [-0.04, 0.42]	0.086	6.3	0.22	0.21 [0.02, 0.39]	**0.032**	9.0	0.16
No	37	172	0.2 [0.12, 0.28]	**0.000**	17.4	0.20	0.21 [0.13, 0.29]	**0.000**	16.6	0.19
Wald test (CWB)				0.954				0.939		
Type of intervention	46	205								
CBT	12	55	0.28 [0.11, 0.45]	**0.007**	6.0	0.18	0.29 [0.17, 0.41]	**0.001**	6.9	0.15
Other	35	150	0.16 [0.08, 0.23]	**0.000**	19.7	0.21	0.18 [0.08, 0.28]	**0.001**	18.0	0.20
Wald test (CWB/HTZ)[b]				0.208				0.145		

Note. Bold degrees of freedom (df) estimates indicate that low df values, which in turn indicate that the variance estimation for the corresponding coefficient is noisy. Following the recommendation of Wu et al. (2025), we do not make statistical significance inference unless the p-value is below 0.01 when $df < 4$. Bold p-values (under the 'p-val'-column) indicate that the coefficients are statistically different from zero at the set .05-level when $df > 4$, whereas italic p-values in the coefficients are statistically different from zero at the set .05-level, when $df < 4$ but failed to be rejected at the adjusted 0.01-level. Est. = Estimate; CI = Confidence Interval; p-val = p-value; Satt. df = Satterthwaite's degrees of freedom; SD = total standard deviation; CWB = Cluster Wild Bootstrap; ITT = Intention-To-Treat; TOT; Treatment-On-the-Treated; QES = Quasi-Experimental Design (non-randomized); RCT = Randomized Controlled Trial; TAU = Treatment as usual. [a] Effects are adjusted for all factors listed in Tables 7, 8, and 9. The exception is the male and number of sessions indicators, as these factors were highly correlated with the sample composition and duration, respectively. Find this information in the PRIMED supplementary material Table 36.

Table S7. Methodologically relevant categorical moderators meta-analysis of social reintegration outcomes

Subgroup-analyses			Unadjusted effects				Covariate-adjusted effects[a]			
Moderator	Studies	Effects	Est. [95% CI]	*p*-val	Satt. df	SD $\sqrt{\tau^2+\omega^2}$	Est. [95% CI] F-statistics	*p*-val	Satt. df	SD $\sqrt{\tau^2+\omega^2}$
Preregistered	46	205								
No	24	65	0.31 [0.21, 0.42]	**0.000**	18.0	0.26	0.36 [0.21, 0.5]	**0.000**	14.6	0.25
Yes	22	140	0.13 [0.03, 0.22]	**0.013**	13.2	0.19	0.14 [0.05, 0.23]	**0.005**	12.9	0.16
Wald test (CWB)				**0.012**				**0.023**		
Test type	46	205								
Clinician-rated measure	12	36	0.13 [-0.08, 0.35]	0.169	4.9	0.22	0.16 [-0.02, 0.33]	0.075	6.5	0.21
Self reported/raw events	39	169	0.21 [0.13, 0.29]	**0.000**	22.6	0.20	0.21 [0.13, 0.29]	**0.000**	16.8	0.17
Wald test (CWB)				0.444				0.430		
Analysis strategy	46	205								
ITT	22	108	0.12 [0.04, 0.21]	**0.007**	12.9	0.20	0.18 [0.08, 0.28]	**0.001**	15.5	0.19
TOT	24	97	0.27 [0.16, 0.38]	**0.000**	10.7	0.19	0.25 [0.13, 0.36]	**0.001**	7.6	0.17
Wald test (CWB/HTZ)[b]				0.056			F(1, 16.5) = 0.96	0.340		
Research design	46	205								
QES	8	18	0.28 [0.01, 0.56]	**0.046**	5.8	0.23	0.20 [-0.09, 0.48]	0.155	10.4	0.11
RCT	38	187	0.18 [0.11, 0.26]	**0.000**	20.4	0.21	0.22 [0.15, 0.29]	**0.000**	18.3	0.18
Wald test (CWB/HTZ)[b]				0.440			F(1, 10.89) = 0.02	0.894		
Control group	46	205								
Individual treatment	3	4	-0.06 [-0.82, 0.7]	0.737	1.8	0.16	-0.15 [-1.1, 0.8]	0.600	**2.3**	0.34
TAU and Waitlist	39	188	0.19 [0.12, 0.27]	**0.000**	21.2	0.20	0.23 [0.15, 0.3]	**0.000**	15.5	0.18
Waitlist only	4	13	0.42 [0.12, 0.72]	*0.024*	**2.5**	0.17	0.24 [-0.48, 0.96]	0.424	4.7	0.29
Wald test (CWB)				0.107				0.542		
Serious/high risk of bias	46	205								
No	31	163	0.2 [0.11, 0.29]	**0.000**	14.3	0.20	0.23 [0.14, 0.31]	**0.000**	15.7	0.19
Yes	16	42	0.2 [0.02, 0.38]	**0.030**	12.4	0.26	0.15 [0.02, 0.29]	**0.030**	6.0	0.12
Wald test (CWB)				0.947				0.415		

Note. Bold degrees of freedom (df) estimates indicate that low df values, which in turn indicate that the variance estimation for the corresponding coefficient is noisy. Following the recommendation of Wu et al. (2025), we do not make statistical significance inference unless the *p*-value is below 0.01, when $df < 4$. Bold p-values (under the '*p*-val'-column) indicate that the coefficients are statistically different from zero at the set .05-level when $df > 4$, whereas italic p-values in the coefficients are statistically different from zero at the set .05-level, when $df < 4$ but failed to be rejected at the adjusted 0.01-level. Est. = Estimate; CI = Confidence Interval; *p*-val = *p*-value; Satt. df = Satterthwaite's degrees of freedom; SD = total standard deviation; CWB = Cluster Wild Bootstrap; ITT = Intention-To-Treat; TOT; Treatment-On-the-Treated; QES = Quasi-Experimental Design (non-randomized); RCT = Randomized Controlled Trial; TAU = Treatment as usual. [a] Effects are adjusted for all factors listed in Tables 7, 8, and 9. The exception is the male and number of sessions indicators, as these factors were highly correlated with the sample composition and duration, respectively. Find this information in the PRIMED supplementary material Table 36. [b] For the covariate-adjusted model, we had a convergence issue when estimating CWB *p*-values. Therefore, we reported HTZ Wald test *p*-values for this model.

Table S8. Substantive continuous moderators meta-analysis of social reintegration outcomes

Moderator	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6[a]
Age	0.003 (0.005)					0.003 (0.005)
% Male		0.133 (0.152)				-0.006 (0.185)
Sessions			*0.021 (0.012)*			*-0.002 (0.015)*
Duration				-0.003 (0.002)		0 (0.002)
Follow-up timing					*0 (0.001)*	*0 (0.001)*
Intercept	0.191 (0.038)***	0.204 (0.037)***	0.190 (0.037)***	0.213 (0.039)***	0.197 (0.033)***	0.204 (0.035)*
Study-level SD	0.072	0.079	0.074	0.056	0.067	0
Effect-level SD	0.193	0.193	0.193	0.194	0.194	0.182
Total SD	0.206	0.208	0.207	0.202	0.205	0.182
Number of effects	205	205	205	205	205	205
Number of studies	46	46	46	46	46	46

Note: **p < .05, **p < .01, ***p < .001.* Italic numbers indicate that the degrees of freedom are below four. SD = Standard Deviation. Cluster robust standard errors in parentheses.

[a] Effects are adjusted for all factors listed in Tables 7 and 8. The exception is the male and number of sessions indicators, as these factors were highly correlated with the sample composition and duration, respectively. Find this information in the PRIMED supplementary material Table 36.

Table S9. Substantive categorical moderators meta-analysis of mental health outcomes

Subgroup-analyses			Unadjusted effects				Covariate-adjusted effects[a]			
Moderator	Studies	Effects	Est. [95% CI]	p-val	Satt. df	SD $\sqrt{\tau^2 + \omega^2}$	Est. [95% CI] F-statistics	p-val	Satt. df	SD $\sqrt{\tau^2 + \omega^2}$
Outcome type	42	144								
Anxiety	9	14	0.4 [-0.07, 0.86]	0.082	7.3	0.56	0.21 [-0.12, 0.54]	0.163	5.7	0.40
Depression	20	37	0.34 [0.09, 0.59]	**0.011**	18.1	0.47	0.27 [0.05, 0.49]	**0.017**	15.0	0.29
General mental health	29	72	0.13 [-0.01, 0.28]	0.074	25.4	0.30	0.25 [0.11, 0.38]	**0.001**	17.5	0.17
Symptoms of psychosis	7	21	0.16 [-0.01, 0.33]	0.055	4.6	0.28	0.21 [0.03, 0.38]	0.029	7.5	0.28
Wald test (CWB)				0.418				0.897		
Schizophrenia in sample	42	144								
Yes	7	14	0.10 [0.04, 0.16]	**0.009**	**3.8**	0.00	0.26 [0.05, 0.47]	**0.022**	6.1	0.00
No	35	130	0.23 [0.09, 0.38]	**0.003**	31.6	0.37	0.24 [0.1, 0.38]	**0.002**	20.9	0.27
Wald test (CWB)				0.100				0.685		
Type of intervention	42	144								
CBT	12	49	0.36 [-0.01, 0.72]	0.054	10.8	0.53	0.33 [-0.03, 0.68]	0.071	10.2	0.44
Other	31	95	0.15 [0.04, 0.26]	**0.008**	23.3	0.20	0.21 [0.11, 0.31]	**0.000**	15.2	0.11
Wald test (CWB)				0.275				0.577		

Note. Bold degrees of freedom (df) estimates indicate that low df values, which in turn indicate that the variance estimation for the corresponding coefficient is noisy. Following the recommendation of Wu et al. (2025), we do not make statistical significance inference unless the p-value is below 0.01, when $df < 4$. Bold p-values (under the 'p-val'-column) indicate that the coefficients are statistically different from zero at the set .05-level when $df > 4$, whereas italic p-values in the coefficients are statistically different from zero at the set .05-level, when $df < 4$ but failed to be rejected at the adjusted 0.01-level. Est. = Estimate; CI = Confidence Interval; p-val = p-value; Satt. df = Satterthwaite's degrees of freedom; SD = total standard deviation; CWB = Cluster Wild Boostrap; ITT = Intention-To-Treat; TOT; Treatment-On-the-Treated; QES = Quasi-Experimental Design (non-randomized); RCT = Randomized Controlled Trial; TAU = Treatment as usual. [a] Effects are adjusted for all factors listed in Tables 10, 11, and 12. The exception is the risk of bias, control group, and number of sessions indicators, as these factors were highly correlated with the research design, age, and duration, respectively. Find this information in the PRIMED supplementary material Table 41.

Table S10. Methodologically relevant categorical moderators meta-analysis of mental health outcomes

Subgroup-analyses			Unadjusted effects				Covariate-adjusted effects[a]			
Moderator	Studies	Effects	Est. [95% CI]	*p*-val	Satt. df	SD $\sqrt{\tau^2+\omega^2}$	Est. [95% CI]	*p*-val	Satt. df	SD $\sqrt{\tau^2+\omega^2}$
Preregistered	42	144								
No	22	62	0.25 [0.03, 0.48]	**0.028**	19.9	0.44	0.35 [0.16, 0.54]	**0.002**	13.6	0.38
Yes	20	82	0.2 [0.06, 0.35]	**0.010**	16.1	0.25	0.21 [0.08, 0.34]	**0.005**	11.7	0.14
Wald test (CWB)				0.696				0.176		
Test type	42	144								
Clinician-rated measure	14	41	0.46 [0.16, 0.76]	**0.006**	12.5	0.46	0.36 [0.12, 0.6]	**0.008**	9.2	0.21
Self reported/raw events	33	103	0.16 [0.03, 0.3]	**0.017**	28.8	0.30	0.22 [0.06, 0.39]	**0.008**	22.8	0.26
Wald test (CWB)				**0.045**				0.212		
Analysis strategy	42	144								
ITT	18	69	0.05 [-0.08, 0.18]	0.401	12.8	0.17	0.16 [0, 0.31]	**0.047**	13.9	0.13
TOT	24	75	0.35 [0.15, 0.56]	**0.002**	21.4	0.41	0.34 [0.14, 0.55]	**0.003**	17.5	0.34
Wald test (CWB)				**0.012**				0.141		
Research design	42	144								
QES	8	17	0.12 [-0.35, 0.58]	0.562	6.6	0.54	0.23 [-0.35, 0.81]	0.379	7.3	0.59
RCT	34	127	0.23 [0.11, 0.35]	**0.000**	28.4	0.26	0.26 [0.17, 0.35]	**0.000**	17.8	0.13
Wald test (CWB)				0.582				0.921		
Control group	42	144								
Individual treatment	2	3	-0.22 [-4.12, 3.67]	0.598	**1.0**	0.41	-0.14 [-9.73, 9.45]	0.894	**1.1**	1.20
TAU and Waitlist	37	134	0.21 [0.1, 0.32]	**0.000**	29.4	0.25	0.3 [0.21, 0.39]	**0.000**	16.3	0.14
Waitlist only	3	7	0.68 [-1.84, 3.2]	0.366	**2.0**	1.03	0.31 [-0.48, 1.09]	0.315	**3.3**	0.41
Wald test (CWB)				0.292				0.931		
Serious/high risk of bias	42	144								
No	30	115	0.24 [0.1, 0.38]	**0.001**	26.3	0.30	0.28 [0.15, 0.4]	**0.000**	17.6	0.13
Yes	13	29	0.17 [-0.12, 0.46]	0.216	11.1	0.45	0.28 [0.1, 0.45]	**0.009**	5.4	0.46
Wald test (CWB)				0.631				0.964		

Note. Bold degrees of freedom (df) estimates indicate that low df values, which in turn indicate that the variance estimation for the corresponding coefficient is noisy. Following the recommendation of Wu et al. (2025), we do not make statistical significance inference unless the *p*-value is below 0.01, when $df < 4$. Bold p-values (under the '*p*-val'-column) indicate that the coefficients are statistically different from zero at the set .05-level when $df > 4$, whereas italic p-values in the coefficients are statistically different from zero at the set .05-level, when $df < 4$ but failed to be rejected at the adjusted 0.01-level. Est. = Estimate; CI = Confidence Interval; *p*-val = *p*-value; Satt. df = Satterthwaite's degrees of freedom; SD = total standard deviation; CWB = Cluster Wild Boostrap; ITT = Intention-To-Treat; TOT; Treatment-On-the-Treated; QES = Quasi-Experimental Design (non-randomized); RCT = Randomized Controlled Trial; TAU = Treatment as usual. [a] Effects are adjusted for all factors listed in Tables 10, 11, and 12. The exception is the risk of bias, control group, and number of sessions indicators, as these factors were highly correlated with the research design, age, and duration, respectively. Find this information in the PRIMED supplementary material Table 41.

Table S11. Substantive continuous moderators meta-analysis of mental health outcomes

Moderator	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6[a]
Age	0.025 (0.007)**					0.023 (0.008)*
% Male		0.262 (0.216)				0.059 (0.243)
Sessions			*-0.061 (0.022)*			*-0.077 (0.045)*
Duration				-0.005 (0.003)		0 (0.003)
Follow-up timing					*-0.001 (0.001)*	-0.001 (0.001)
Intercept	0.213 (0.054)***	0.231 (0.063)***	0.235 (0.062)***	0.249 (0.073)**	0.220 (0.064)**	0.257 (0.078)*
Study-level SD	0.231	0.301	0.283	0.295	0.299	0.213
Effect-level SD	0.151	0.15	0.15	0.15	0.15	0.152
Total SD	0.275	0.337	0.32	0.331	0.335	0.261
Number of effects	144	144	144	144	144	144
Number of studies	42	42	42	42	42	42

Note: *$*p < .05$, $**p < .01$, $***p < .001$.* Italic numbers indicate that the degrees of freedom are below four. SD = Standard Deviation. Cluster robust standard errors in parentheses.

[a] Effects are adjusted for all factors listed in Tables 10 and 11. The exception is the risk of bias, control group, and number of sessions indicators, as these factors were highly correlated with the research design, age, and duration, respectively. Find this information in the PRIMED supplementary material Table 41.

Detailed overview of included outcomes

In this section, we describe all the included outcome measures. In the protocol, we describe that we do not clearly distinguish between primary and secondary outcomes. However, as all of the outcomes highlighted in the protocol can be said (directly or indirectly) to belong to social reintegration, we consider this to be our primary outcome. Meanwhile, studies that assess social reintegration outcomes commonly include measures of mental health as well. Originally, these types of outcomes were not the main focus of the review. Therefore, we consider mental health outcomes to represent secondary outcomes. Below, one can find a full list of all included outcome measures and scales.

Primary outcomes: social reintegration

The relevant outcomes for this review broadly concern problem behaviors and social difficulties associated with social marginalization. Broadly, we characterized these types of outcomes as reintegrational outcomes where we interpret a decrease on the given scale to be an indirect indicator of increased reintegration into society. The specific outcomes include:

- Alcohol/substance abuse
- Self-harming behaviour
- Criminal behaviour
- Loneliness
- Self-efficacy
- Homelessness
- Poverty
- Unemployment
- Hospital admissions
- Participants' subjective well-being and quality of life

In order for a study to be included, the study needed to include at least one measure of social reintegration. Below, we list the exact outcomes that were included in the review:

Social functioning (impairment)

- Areas of Change Index (ACI)
- Clinician-rated Global Assessment of Functioning Scale (GAF)
- Global Assessment Scale (GAS)
- Global Assessment of Functioning scale (GAF)
- Inventory of Interpersonal Problems 64
- Life Skills Profile 16 (LSP-16)
- Global Assessment of Functioning Scale (GAF)
- Personal and Social Performance Scale (PSP)
- Social Adaptation Self-Evaluation Scale (SASS)
- Social Function Questionnaire
- Sheehan Disability Scale (Functional status)

- The Disability Assessment Schedule (DAS-II)
- The Social Skills Performance Assessment (SSPA)
- Satisfaction with changes in overall daily functioning (single-item self-report: SCI)

Loneliness

- The Loneliness Scale
- 11-item De Jong Gierveld scale,
- The UCLA Loneliness Scale

Hope, Empowerment & Self-efficacy

- 13-item General Help-Seeking Questionnaire
- 6-item Job Search Self-Efficacy Scale
- Beck's Hopelessness Scale
- The Core Components of Treatment Scale
- Coping with Stress Self-efficacy (CSSE)
- The Dutch Empowerment Scale
- The Empowerment Scale
- The Herth Hope Index
- The Integrative Hope Scale (IHS)
- The Recovery Assessment Scale
- The Self-Efficacy Scale (SES)
- The Self-Identified Stage of Recovery Scale
- The Work Hope Scale, The Work Motivation Scale
- Miller Hope Scale (MHS)
- Morton et al. (2012) The Beck Hopelessness Scale
- Netherlands Empowerment List
- Rogers Empowerment Scale (RES)
- Rosenberg Self-Esteem Scale (RSES)

Subjective Wellbeing and Quality of Life

- General Wellbeing Scale Five-item EQ-5D
- Physical health related quality of life; Mental health related quality of life
- PSYCHOLOPS (consists of four questions, three domains: problems, functions, and well-being)
- Quality of Life Enjoyment and Satisfaction Questionnaire (Q-LES-Q)
- Quality of Life: Short Form Health Survey (SF-36)

- Retrospective quality of life scale (RQOL)
- Short-Form Health Survey (SF-36)
- The Manchester Short Assessment of Quality of Life
- The Perceived Stress Scale (PSS)
- The Quality of Life Scale
- The Quality of Life index
- The Warwick Edinburgh Mental Wellbeing Scale (WEMWBS)
- WHO Quality of Life-BREF

Self-esteem (positive views of the self, decreased self-stigma)

- 5-item self-concurrence subscale of the Self-Stigma of Mental Illness Scale-Short Form
- Internalized Stigma of Mental Illness scale
- Link Perceived Stigma Questionnaire (LPSQ)
- Modified Engulfment Scale (MES)
- Tennessee Self-Concept Scale (TSCS)
- The Rosenberg Self-Esteem Scale (RSES)
- The State Self-Esteem Scale (SSES)

Homelessness

- Housing Status in Past 60 days

Unemployment

- Employment outcome (self-reported)
- The Employment and Vocational Activities Checklist

Alcohol and Substance Abuse

- Adaptation of the Addiction Severity Index plus Saliva testing
- Alcohol Use Disorder Identification Test
- Drug Abuse Screening Test
- Opiate Treatment Index
- Substance Use in Past 30 days
- The Addiction Severity Index
- The Addiction Severity Index-lite (ASI-lite)
- The Alcohol Use Disorders Identification test (AUDIT)
- The Severity of Dependence Scale

Secondary outcomes: mental health

As long a the studies had a broad focus on social integration beyond reduction of mental health symptoms, we extracted psychiatric symptoms/mental health measures as secondary outcomes when these were reported. Specifically, we extracted the following mental health outcome:

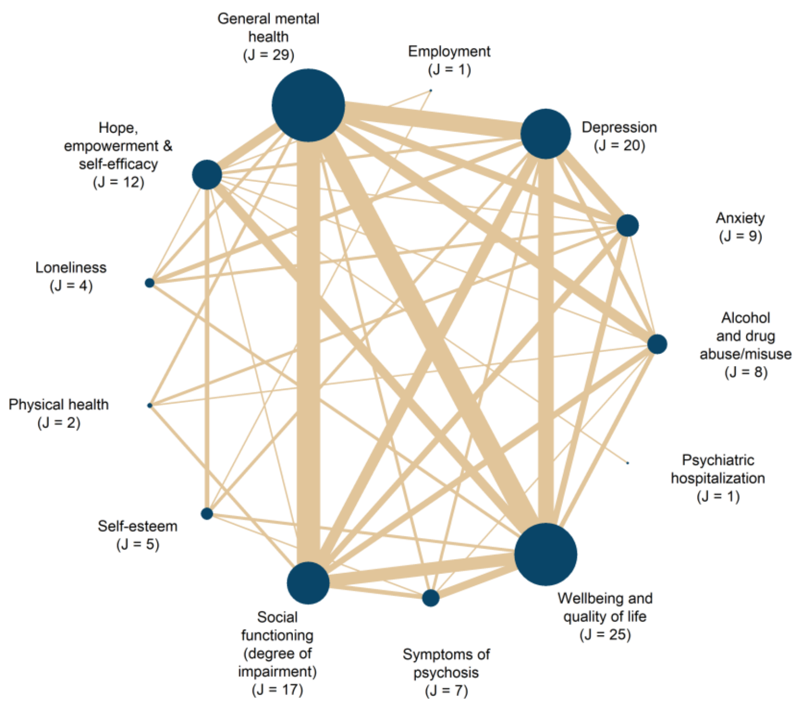
General mental health, Anxiety, Depression, and Symptoms of Psychosis.

- Beck's Anxiety Inventory (BAI)
- Beck's Depression Inventory (BDI)/Beck Depression Inventory II (BDI-II)
- Brief Assessment of Cognition in Schizophrenia
- Brief Psychiatric Rating Scale (BPRS)
- Brief Symptom Inventory (BSI)
- Calgary Depression Scale for Schizophrenia (CDSS)
- Clinical Global Impression Scale for Bipolar Disorder
- Clinical Global Impression Scale (CGI)
- Clinical Outcomes in Routine Evaluation Outcome Measure (CORE-OM)
- Current Mental Health Symptoms
- Depression, Anxiety, and Stress Scale-21 (DASS 21)
- The Depression Anxiety Stress Scale (DASS)
- The Difficulties in Emotion Regulation Scale (DERS)
- The Expanded Brief Psychiatric Rating Scale
- The Hamilton Depression Rating Scale (HAM-D), BDI
- The Hopkins Symptoms Checklist (HSCL-25)
- The Liebowitz Social Anxiety Scale
- The Mental Health Confidence Scale
- The Mini Social Phobia Inventory
- The Modified Scale for the Assessment of Negative Symptoms
- The Overall Anxiety Severity and Impairment Test (OASIS)
- The Patient Health Questionnaire (PHQ-9)
- Perceived Stress Scale (PSS)
- The Positive and Negative Syndrome Scale (PANSS)
- The Recovery Assessment Scale
- The 15-item German version of the Center for Epidemiologic Studies-Depression Scale (CES-D)
- The Scale for the Assessment of Negative Symptoms (SANS)
- The Scale for the Assessment of Positive Symptoms (SAPS)
- The Short Borderline Symptom List (BSL-23)
- The Symptom Checklist 90-Revised (SCL-90-R)
- The Toronto Alexithymia Scale – 20 (TAS-2)
- The PTSD Symptom Scale Interview (PSS-I)
- The Borderline Evaluation of Severity over Time (BEST)
- The Young Mania Rating Scale (YMRS)
- Self-reported recovery
- The Brief Fear of Negative Evaluation Scale
- Short version of the Social Phobia Inventory (mini-SPIN)

- The Positive and Negative Syndrome Scale
- The 53-item Brief Symptom Inventory
- The Anxiety Disorder (GAD-7) measure
- Personal Health Questionnaire Depression Scale (PHQ-9)

To understand the interconnectivity between all the above-listed outcomes, Figure S6 maps the interconnections between all reported outcomes.

Figure S6 | Network structure of contrasts between primary and secondary outcome constructs



Search methods for identification of studies

Relevant studies were identified through electronic searches of bibliographic databases, governmental and grey literature repositories, hand searched in specific targeted journals, attempts to contact experts, and Internet search engines. The electronic database searches were completed in September 2022, and other resources were completed in July 2024. We searched to identify both published and unpublished literature. The searches were international in scope. Reference lists of included studies used in the meta-analysis were also searched.

We implemented a wide range of search methods and strategies to maximise coverage of relevant references, while simultaneously attempting to reduce different types of bias related to publication and dissemination systems.

Subject terms in the facets were selected according to the thesaurus or index of each database. Keywords were supplied if the search technique provided additional results. Use of truncation and wildcards was used to address English spelling variants.

The different strategies and methods are presented below, and detailed documentation of the searches is available upon request.

Electronic searches

The following databases were searched electronically:

MEDLINE (OVID) 1966 - 2022

EMBASE (OVID) 1974 - 2022

APA PsycINFO (EBSCO) 1800 - 2022

CINAHL (EBSCO) 1981 - 2022

Sociological Abstracts (ProQuest) 1952 - 2022

Social Services Abstracts (ProQuest) 1979 - 2022

SocINDEX (EBSCO) 1908 - 2022

Academic Search Premier (EBSCO) 1975 - 2022

International Bibliography of the Social Sciences (IBSS) (ProQuest) 1951 - 2022

Science Citation Index (Web of Science Core Collection) 1900 - 2022

Social Sciences Citation Index (Web of Science Core Collection) 1990 - 2022

Cochrane Central Register of Controlled Trials (CENTRAL) (1996) – 2022

Searching other resources

In addition to electronic databases, we also searched the following venues:

- Google Scholar—https://scholar.google.com
- Google —https://www.google.com/
- Searches in Google and Google Scholar
- Social Science Research Network — https://papers.ssrn.com/sol3/DisplayAbstractSearch.cfm
- CORE – https://core.ac.uk Internationale repositorier
- Danish National Research Database—http://www.forskningsdatabasen. dk/en
- NORA—Norwegian Open Research Archives—http://nora. openaccess.no/
- Cristin - Current Research Information SysTem In Norway - https://wo.cristin.no/as/WebObjects/cristin.woa/wa/fres?la=no
- SwePub—Academic publications at Swedish universities—http:// swepub.kb.se/
- DIVA - https://www.diva-portal.org/smash/search.jsf?dswid=69

Searches for working papers and conference proceedings in English

SHS Web of Conferences (www.shs-conferences.org) Open Access proceedings in Humanities and Social Sciences

The Social Care Institute for Excellence (SCIE) : www.scie.org.uk/publications/index.asp

Searches for Government Documents

NICE National Institute for Health and Care Excellence www.nice.org.uk

Searches for Dissertations

EBSCO Open Dissertations (https://biblioboard.com/opendissertations/)

Open Access Theses and Dissertations (oatd.org)

Hand Searches

Hand searches were carried out in selected journals:

- BMC Public Health
- BMC Psychiatry
- Journal of Psychosocial Rehabilitation and Mental Health
- Psychiatric Quarterly
- Community Mental Health Journal
- Disability and rehabilitation
- International journal of mental health systems

- Sociology of health and illness

Searches included all issues published in 2022, 2023, and up to May 2024

Citation-tracking

We also checked the references for all identified existing systematic reviews and meta-analyses and all included primary studies.

Contacting experts in the field

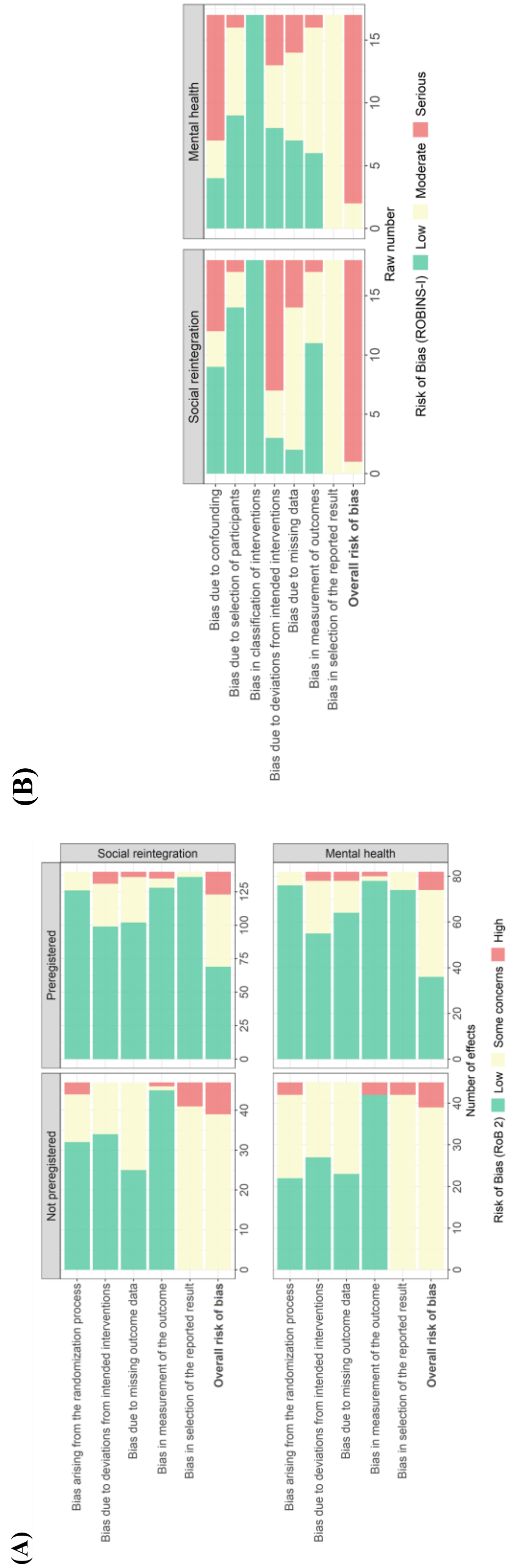
We did not contact international experts.

Language restrictions

We reviewed studies published in English, Danish, Swedish, and Norwegian.

Risk of bias plot

Figure S7. Risk of bias assessments of RCTs and non-randomized studies.



Note. Panel (A) is based on 187 social reintegration effect sizes from 38 studies and 127 mental health effect sizes from 34 studies. Panel (B) is based on 18 social reintegration effect sizes from 8 studies and 17 mental health effect sizes from 8 studies. studies.

Treatment measurement and effect size calculation

Further details about the effect size calculation are given below.

All effect size calculations were conducted using R 4.5.1 and R 4.5.2[2] and RStudio[3]. Specifically, we drew substantially on the `tidyverse` packages[4] for data manipulation, visualization, and summary statistics of various kinds. We calculated effect sizes from medians and quantiles by using the `estmeansd`[5] (version 1.0.1), and we used the `metafor` package[6] (version 4.0-8) to aggregate within-study effect sizes not relevant to our moderator analyses.

Moreover, all effect size calculations can be found in the Online Appendix A. In this scheme, we have calculated effect sizes individually for each study. This means that one can press on each study to see the exact methods used to calculate effect sizes for the particular study. We developed this scheme in response to widespread critiques indicating that effect sizes reported in (psychological) meta-analyses are generally difficult to reproduce[7].

Effect size calculation

We calculated SMDs using the Hedges' *g* estimator, which corrects for the small-sample bias inherent in Cohen's *d*.[8] For sensitivity analyses, we also calculated Cohen's *d*. All SMDs were coded so that a positive value indicated a beneficial effect of the group-based treatment. Accordingly, effect sizes were reversed for test scales where lower scores indicated better outcomes (e.g., improved condition). For an illustration of this procedure, see Lim et al.[9]. Across all interventions and outcomes, we calculated 349 SMD estimates clustered within 49 studies. All in all, 205 effect sizes from 46 studies captured social reintegration outcomes, while 144 effect sizes from 42 studies captured mental health outcomes. All but two effect sizes represent pretest-adjusted effect sizes.

As we computed effect sizes from the result data deduced from various research designs, estimation techniques, and reporting standards, we applied a wide range of different methods to obtain the relevant statistics for effect size calculation[10–16]. Specifically, to increase the internal validity and precision of Hedges' *g*, we prioritized calculating pretest-/baseline- and/or covariate-adjusted versions of the *g* metric[12,14,16,17] . As a further attempt to increase the statistical power of our analyses, we reduced (artificial[1]) within-study variability by aggregating all study results that were reported across subgroups that were not pre-specified in our protocol, as recommended by Vembye, Pustejovsky et al.[18]. Specifically, this aims to mitigate *chance bias* (i.e., that the effect is primarily driven by randomization not adequately balancing all covariates, which often happens in small studies, or when many outcomes are reported) to induce large impacts on the true variation measures. See Sacks et al.[19] for an example of this procedure.

Although participants in the majority of studies had been individually randomized to treatment and control groups, the fact that the intervention was provided in a group format at the same time and space

[1] Within-study variability can be artificially inflated when small sample studies report large amounts of effect sizes, as sample error (i.e., chance bias) in this case can make the variability look more extreme than it truly is (see Vembye, Pustejovsky et al.[18])

creates dependence among members of the same group, as they share common traits such as receiving treatment from the same therapist/professional, etc. If not accounted for, this yields effect size standard errors that are incorrect. In this regard, we followed the recommendation from the Cochrane Handbook (section 23.1.8[13]) to adjust for clustering arising from this type of clustering caused by group treatment.

Therefore, to ensure comparability between effect sizes[20], all effect sizes and the corresponding variance estimates were standardized by the *total variance*. That is, we computed effect sizes and variance estimates that incorporate both the variation arising from the participant/individual level as well as the cluster level, which means the group-based treatment level. To do so, we cluster-bias adjusted all effect sizes from studies, ignoring the nesting of participants in the given group-based treatment. Only two of the included studies accounted for this issue in the statistics we used for effect size computation (i.e., Haslem et al.[21]; Michalak et al.[22]).[2]

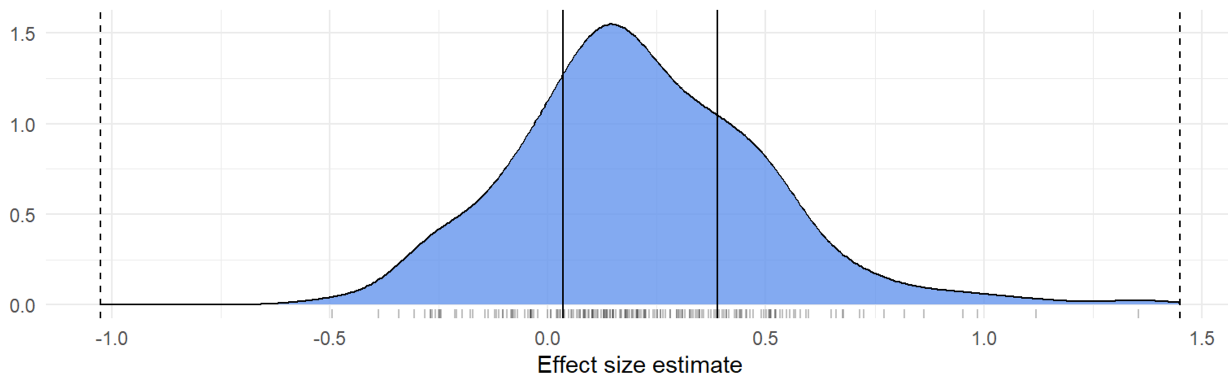
In this review, all studies represent so-called *partially clustered* studies, with clustering arising in the treatment group only. Therefore, we used the cluster-bias methods developed by Hedges and Citkowicz[25] that specifically account for this design issue. Because our priority was the covariate-adjusted effect size, and given that Hedges and Citkowicz primarily developed cluster formulas for posttest-only study designs, we integrated their formulas with those presented by WWC[16] and Hedges et al.[12].[3] Vembye[26] provides an overview of the exact formulas used can be found at https://mikkelvembye.github.io/VIVECampbell/reference/vgt_smd_1armcluster.html.

A common challenge with cluster-bias adjustments is that they are premised upon intraclass correlation (ICC) values, which are rarely reported in practice. Among the included studies, only three studies (Crawford et al.[24]; Haslem et al.[21]; van Gestel-Timmermans et al.[23]) reported ICC values; otherwise, ICC values were imputed, as suggested by Hedges[27] (2007). We imputed ICC values of 0.1 for the main analyses and conducted sensitivity analyses imputing ICC equal to 0.05 and 0.2. For further details, see section 'Unit of analysis issues' below.

The final distributions of effect sizes for social reintegration and mental health outcomes, respectively, are presented in Figures S7-S8.

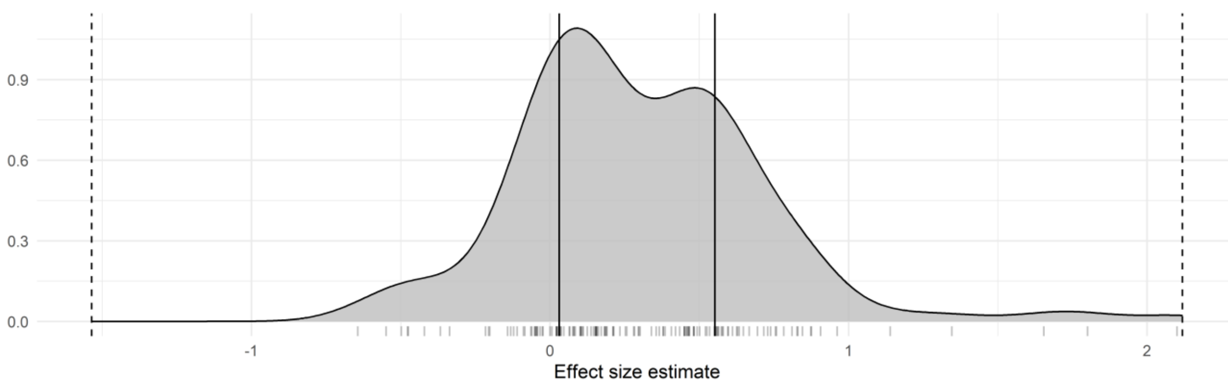
[2] van Gestel-Timmermans et al.[23] and Crawford et al.[24] accounted for the multi-level structure of their data, but we were not capable/unsure of how to calculate effect size from the reported model results.
[3] For this matter, we consulted Larry Hedges, who confirmed the appropriateness of this approach.

Figure S8. Empirical distribution of social reintegration (primary outcome) effect size estimates



Note: Empirical distribution of social reintegration effect size estimates. Solid vertical lines indicate lower and upper quartiles. Dashed lines indicate the 1st quartile minus 3 times the inter-quartile range and the 3rd quartile plus 3 times the interquartile range. Effect sizes outside of the range of dashed lines would be considered outliers according to Tukey's[28] definition.

Figure S9. Empirical distribution of mental health (secondary outcome) effect size estimates



Note: Empirical distribution of mental health effect size estimates. Solid vertical lines indicate lower and upper quartiles. Dashed lines indicate the 1st quartile minus 3 times the inter-quartile range and the 3rd quartile plus 3 times the interquartile range. Effect sizes outside of the range of dashed lines would be considered outliers according to Tukey's[28] definition.

For more detailed visualization across more specific outcomes, see the Online Appendix B, Figures 100-102.

Further technical description of the effect size calculation

To describe the above procedure more formally, the Hedges' g estimator we used can be written as

$$g_t = \kappa \times \left(\frac{b}{S}\right) \times \gamma \tag{1}$$

The subscripted t in (1) indicates that the g metric was standardized by the total variance. κ is the small study corrector, $1 - \frac{3}{4*df-1}$, where df is the degrees of freedom, which typically equals N . That is the total sample size of the study. Yet, as all included studies are partially cluster-designed studies, we calculated cluster-adjusted degrees of freedom (df_{cl}) as (c.f. Hedges & Citkowicz[25], Equation 7):

$$df_{cl} = \frac{[(N-2)(1-\rho_{ICC}) + (N_T - n)\rho_{ICC}]^2}{(N-2)(1-\rho_{ICC})^2 + (N_T - n)n\rho_{ICC}^2 + 2(N_T - n)(1-\rho_{ICC})\rho_{ICC}} \quad (2)$$

Where N is given as above, ρ_{ICC} is the intraclass correlation, N_T is the sample size of the treatment group, and n is the average cluster size. That is the average sample size of the group-based treatment format. As previously described, we could only obtain empirical values of ρ_{ICC} from three studies. Thus, we most often imputed this value. Whenever possible, we empirically obtained values of n from the included studies. If the average sample size of the group treatment was not reported, we assumed $n = 8$. This value equals the median group size for studies that empirically reported this value and reported social reintegration outcomes.

Next, b in Equation (1) denotes the mean difference between the group-based treatment group and the individual control group. Most commonly, we computed b as a difference-in-differences estimate, that is, $(M_T^{post} - M_T^{pre}) - (M_C^{post} - M_C^{pre})$, where, M_T^{pre}, M_C^{pre}, M_T^{post}, and M_C^{post} represent the pre- and posttest means for the treatment and control groups, respectively. We define this type of effect size as a difference-in-difference (DID) effect size. For 40 studies, we calculated the nominator of Equation (1) from the raw pre- and posttest means.

For five studies (van Gestel-Timmermans et al.[23]; Gonzalez & Prihoda[29]; McCay et al.[30]; Smith et al.[31], 2021; Wuthrich & Rapee[32]), b was obtained either from repeated ANOVAs or estimated marginal means. Moreover, b was obtained as regression estimates from multi-level regression models for two studies (Haslam et al.[21]; Michalak et al.[22])[4]. One study (Gutman et al.[33]) provided the raw data. From this data, we estimated pretest-adjusted effects using standardized linear regression. Finally, for one study only (Bond et al.[34]), we calculated the posttest-only version of b, that is $b = (M_T^{post} - M_C^{post})$.

S in Equation (1) represents the pooled standard deviation (SD). Sometimes referred to as the pooled within-group SD[35]. This was most frequently calculated (i.e., for 318 effect sizes and 43 studies) as $S =$

[4] Note, from Michalak et al.[22], we 4 effect size from the raw means and 10 effect size from multi-level models. The standard deviation used to standardize the mean effect difference also varied within this study.

$\sqrt{\frac{(N_T-1)SD_T^2+(N_C-1)SD_C^2}{N_T+N_C-2}}$, where N_C is the sample size of the control group, whereas SD_T and SD_C represent the SD of the treatment and control group, respectively. Four studies (Haslam et al.[21]; Smith et al.[31]; Wojtalik et al.[36]; Wuthrich & Rapee[32]) did not report posttest SDs. In these cases, we used the pretest standard deviation instead.

For one study (McCay et al.[30]), where we extracted Cohen's *d* estimates calculated by the authors, it was unclear how S is calculated, but we assumed that a pooled SD was used as prescribed for Cohen's d. From two studies (Gonzalez & Prihoda[29]; Michalak et al.[22]), we extracted the total SD and not the pooled within-study SD. To align this SD measure, we used Equation E.14 from the What Works Clearinghouse's standard and procedure handbook[35] (version 5, p. 168) to convert them to the pooled within-group SD.

For effect sizes based on the GAF (Global Assessment of Functioning), PHQ-9 (Patient Health Questionnaire), and BDI (Beck Depression Inventory) scales, we calculated S as a population-based standard deviation, following the recommendation of Fitzgerald and Tipton[11]. Specifically, S was estimated by pooling all control group standard deviations. We adopted this approach to increase the generalizability/external validity of the results, as the population-based version of S provides an estimate closer to the population we want to generalize to. We applied this method only to these three scales, as each was reported in at least five studies—the minimum number required for adequately estimating population-based standard deviations[11].

Finally, γ in Equation (1) is the small number of clusters adjustment factor[16] that is computed as $1 - \frac{(N_C+n-2)\rho_{ICC}}{N-2}$ with all quantities given as previously defined.

Next, a general formula expressing the sample variance (V_{gt}) of our Hedges' *g* estimator can be written as

$$V_{gt} = W \times \xi + P \tag{3}$$

here W is the scaled/standardized sampling variance of b from Equation (1), i.e., $\left(\frac{se_b}{S}\right)^2$, which expresses the contribution of the variability of b, whereas P "captures the contribution of the variance of [S]—that is, how precisely estimated is the *scale* [*/standard deviation*] of the outcome"[37]. ξ either represents the small number of clusters adjustment factor, γ, or the design effect, η, depending on whether a study adequately accounted for clustering or not. η is given by $1 + \left(\frac{nN_C}{N} - 1\right)\rho_{ICC}$.

Again, as we extracted data from various research designs, estimation techniques, and reporting standards, we computed W from Equation (3) in several ways. As we most frequently calculated difference-in-differences (DID) effect sizes, we obtained W as $2(1-r)\left(\frac{1}{N_T} + \frac{1}{N_C}\right)$, where r is the pre-posttest correlation. Although r is usually not reported in primary studies, this can be calculated from commonly reported measures. For nine studies[23,33,38–44], we could obtain r directly from the study or calculate r either using the equation from the Cochrane Handbook[13] (section 6.5.2) or Equation 31 from Wilson[15]. As described by Pustejovsky[14], an equivalent alternative to estimating W for covariate-adjusted effect sizes is to use t of F test values, thus $W = \frac{g_t^2}{F} = \left(\frac{g_t}{t}\right)^2$. We used either t or F values from seven studies[22,40,45–49] reporting difference-in-difference and one study[29] reporting repeated ANOVA estimates. If either the *t/F* test seemed flawed or yielded a larger variance estimate relative to the DID variance estimator, we used the DID variance estimator and imputed r when necessary (see Druss et al.[50]; James et al.[51]).

On this matter, Hedges et al.[12] (2023) suggest using test-retest reliability values of a given outcome scale whenever it is impossible to compute r. We imputed test-retest reliability measures as proxies for r in two studies (Cano-Vindel et al.[52]; Morton et al.[53]). For studies where we were not able to apply any of the above methods, we imputed $r = 0.5$.[5] Thus, W reduces to $\left(\frac{1}{N_T} + \frac{1}{N_C}\right)$, which equals W computed for posttest-only effect sizes.

For three studies, we calculated $W = \left(\frac{se_b}{S}\right)^2$, where se_b is the standard error of b, typically, representing the covariate-adjusted mean difference between the treatment and control group. In three studies using ANCOVA-like estimation (i.e., repeated ANOVA and estimated marginal means) methods (McCay et al.[30]; Smith et al.[31]; Wuthrich & Rapee[32]), we estimated $W = (1 - r^2)\left(\frac{1}{N_T} + \frac{1}{N_C}\right)$.

For all covariate-adjusted effect size estimates, we calculated P in Equation (3) as $\frac{g_t}{2(df-q)}$, g_t and df are defined in Equations (1) and (2), and q is the number of covariates. For posttest-only effect sizes $q = 0$, whereas for pretest/baseline-only adjusted effect sizes $q = 1$. We did not adjust V_{gt} for small sample bias (i.e., multiplying κ^2 with Equation 3), as Hedges et al.[12] recommend not doing so.

Further details about the data synthesis

Unit of analysis issues

The unit of analysis of this review was the effects of group-based intervention on the individual participants. However, the fact that group-based interventions are provided to the participant in a group format at the same time and space likely creates dependence/clustering among/of participants. That is, the treatment effects for participants in the same group are more similar compared to other individuals

[5] The raw average pre-posttest correlations (r) for reintegrational and mental health outcomes were .623 and .510, respectively.

who are not a part of the particular group. This breaks the classical statistical assumption of independence[54]. Thus, if this type of clustering is not accounted for, the average individual treatment effects will appear to be more certain than they actually are (i.e., the standard errors will be underestimated). As mentioned in the previous section, we, therefore, cluster-biased corrected all studies to account for this dependence of participants from the same group, which in this case is only an issue in the treatment group.

Although cluster-bias correction of treatments received in groups is recommended by Higgins, Eldridge, et al.[13], "Weiss et al. (2016) indicate that both adjusting and not adjusting are likely to yield biased standard errors in primary studies (over- and underestimated, respectively)"[55]. Therefore, to accommodate this issue, we conducted a sensitivity analysis, using a non-cluster-adjusted version of the g estimator. That is, we did not add the γ and ξ factors to Equations (1) and (3), respectively.

As the statistical method we used for cluster adjustments[25] has not been implemented in any standard software, we developed our own R package `VIVECampbell`[26], in which we implement cluster-bias adjustments when there is clustering in one treatment group only across various effect size estimation techniques.

Other unit of analysis issues

Our protocol states that we intended to analyze posttest (i.e., effects measured one year or less after the end of the intervention) and follow-up (i.e., effects measured more than 1 year after the end of the intervention) effects separately to avoid unit-of-analysis errors. However, we did not detect any follow-up effects. As an exploratory sensitivity analysis, we investigated whether the measurement timing within the first year could explain differences between effect sizes.

Criteria for the determination of independent findings (dependent effect sizes)

As the majority of the included studies contribute multiple effect sizes, the final datasets contain dependencies among the computed effect sizes. The primary dependency structure we detected in our datasets pertained to the *correlated effects dependency structure*, where studies report multiple outcome results from the same or partially the same sample of participants. Most commonly, studies (30 for social reintegration outcomes and 36 for mental health outcomes) reported results for the same participants across different outcomes and/or time points. As noted in Table 1 in the paper, three studies also created correlated dependencies by comparing multiple treatment groups to a single control group.

We did not have any studies creating the so-called *hierarchical effects dependency structure*, in which the dependency arises from outcomes reported across distinct, non-overlapping samples. Finally, 12 (social reintegration) and 8 (mental health) studies reported a single effect size only.

This type of complex dependency requires advanced meta-analytical techniques to avoid the production of overly optimistic results. To address this, we used the *correlated hierarchical effects* (CHE) model family (Pustejovsky & Tipton, 2025). These models account simultaneously for both hierarchical and correlational dependence in meta-analytic data. To guard against model misspecification, we employed either robust variance estimation[56,57] (RVE) or cluster bootstrap methods[58–60]. RVE was used when estimating standard errors for single coefficients in the main analyses, whereas bootstrap methods were applied for Wald test statistics and for publication bias analyses. These applications are described in more detail in later sections.

A challenge when working with dependent effect size data is that the true dependence is largely unknown, requiring the meta-analytist to make arbitrary guesses about the true correlation among the dependent effect size estimates. For the three multi-treatment, however, we were able to asymptotically estimate the correlation among the effect sizes using the `vcalc` function from the `metafor`, which implements formulas from Gleser and Olkin[61] and Wei and Higgins[62]. For all other cases, we assumed a constant within-study correlation of 0.8, as specified in our protocol.

To capture this combination of empirically estimated and assumed correlations, all fitted models are denoted with the prefix *PE* (Partially Empirical), following Pustejovsky & Tipton[63]. For example, the CHE model is denoted *PECHE*, standing for *Partially Empirically Correlated Hierarchical Effects*.

To provide a more profound understanding of the dependency structure of our data, we applied the 'preliminary data analysis for meta-analysis of dependent effect sizes' (PRIMED) workflow.[64] On the one hand, this helped us to illustrate and visualize both the hierarchical and correlational structure of the dataset, with effect sizes nested within studies and with studies reporting on multiple eligible outcomes. On the other hand, it helped us detect any coding errors that were not caught during our quality checks. All the PRIMED analyses can be found in the Online Appendix B at https://osf.io/s2j9a/files.

Dealing with missing data

According to Pigott[65], missing data in meta-analyses arises for three main reasons: (1) missing studies, meaning that some studies cannot be detected for various reasons; (2) missing effect sizes within a study, for example, because certain outcomes are not reported or because statistical measures needed to calculate effect sizes are unavailable; and (3) missing predictor variables, that is, study, sample, or outcome characteristics that researchers wish to use to predict differences in effect sizes but which are not reported. As reasons (1) and (2) primarily reflect publication and reporting biases, we describe how these issues are addressed in the section entitled *'Assessment of reporting biases'* below. A further type of missingness pertains to attrition of participants in the study sample. The consequences of this type of issue are assessed in our risk of bias assessment.

For the latter issue (3), concerning missingness on the predictor variables described in our protocol, we used mean imputation to recover missing values. Although this was not prespecified in our protocol and is not considered to be a state-of-the-art management of the missingness, we did so because we experienced having a maximum of one missing study across all moderator variables prescribed in the protocol. Specifically, we could not obtain information about the average number of males in the study sample for Gordon et al.[47], prompting us to impute means on the 'average percent males in sample' variable related to three social reintegration effect size estimates and one mental health effect size estimate, respectively. Further, we could not back out the number of intervention sessions for Somers et al.[44], causing us to impute means on the 'total number of sessions' variable related to four social reintegration effect size estimates and one mental health effect size estimate, respectively.

As the number of missing values was so few, we found it unnecessary to use more sophisticated techniques such as multiple imputation. This would moreover unnecessarily complete the reliable computation of the cluster robust standard errors and the embedded degrees of freedom (see Vembye et al.[66], for a description of this problem).

Assessment of heterogeneity

We primarily assessed heterogeneity with the measures of between-study (τ) and within-study (ω) SDs, along with the total variation between true effects. That is $\sigma_T = \sqrt{(\tau^2 + \omega^2)}$. We used the restricted maximum likelihood versions of τ and ω. For the overall average effect size estimations, we reported I^2.[67] However, this should not be interpreted as a key indicator of heterogeneity, as this is just a measure of the ratio between the true variation between effects and the sampling error[68]. For the overall average effect size estimates, we also report Q statistics and 67% prediction intervals. With inspiration from Treves et al.[69], we chose to use a 67% prediction interval because this interval covers the most typical range of observations that one would expect to see in a new study. In fact, it represents 2/3 of the most likely effect sizes expected to be seen in a new study.

Overall average effect – PECHE-RVE

As we experienced various types of dependencies in our data, we derived the overall average effects for social integrational and mental health outcomes, using the partially empirical correlated-hierarchical effects (PECHE-RVE) model[63], as this model can handle various types of dependencies among effect sizes. Specifically, this model accounts for the hierarchical data structure, with effect sizes nested within studies, allowing us to obtain heterogeneity measures at both the study-level and the effect size-level. This provides valuable diagnostic information about at what levels unexplained heterogeneity might be explained. We used restricted maximum likelihood to estimate the between-study (τ^2) and within-study (ω^2) heterogeneity[70,71].

Moreover, the model accounts for correlation among the effect size standard errors, either by imputing or estimating the covariance between within-study effect sizes. In our data, we were able to estimate the covariance for some within-study effects. However, for the majority of the effect sizes, we estimate the study's variance-covariance matrix by imputing $\rho = 0.8$, as specified in our protocol.

As previously described in the 'Criteria for the determination of independent findings' section, we used robust variance estimation (RVE) to guard against model misspecifications.

Further model details and notation for PECHE-RVE

To explicate the statistical models used in this review, consider that our data represent a collection of J studies, each contributing $k_j \geq 1$ effect size estimate. Let T_{ij} be the effect size estimate i ($i = 1 \dots, k_j$) from study j ($j = 1, \dots, J$) with known and fixed sampling variance s_{ij}^2. Then let x_{ij} denote a row vector of p covariates and $\boldsymbol{\beta}$ denote a vector of p regression coefficients. In formal parlance, the PECHE working model can be written as:

$$T_{ij} = x_{ij}\boldsymbol{\beta} + u_j + v_{ij} + e_{ij} \tag{5}$$

For the intercept-only model, estimating the overall average effect, x_{ij} reduces to a row vector of 1's and $\boldsymbol{\beta}$ represents the overall average effect. In this model, it is assumed that the study-level and within-study random effects follow and normal distribution with $u_j \sim N(0, \tau^2)$ and $v_{ij} \sim N(0, \omega^2)$. Also, the sampling errors $e_{ij} = T_{ij} - \boldsymbol{\beta}$ are assumed to follow a normal distribution with $e_{ij} \sim N(0, s_{ij}^2)$.

To account for correlated dependency, the model assumes that all within-study effects are equally correlated, thus that $Cov(e_{ij}, e_{hj}) = \rho s_{ij} s_{hj}$ for effect sizes $i \neq h$ within study j. As the correlation ρ among within-study effect sizes is usually unknown, it the often imputed and assumed to be constant across studies. However, when studies included multiple treatment groups compared to the same control group, the covariance between within-study effects can be asymptotically approximated. We used this approach when estimating the variance-covariances for the three studies with multiple treatment groups. We thereby used a mixture of empirically known and unknown covariance estimates.

Subgroup analysis and investigation of heterogeneity

To investigate whether focal moderators could explain the true difference in effect sizes, we conducted a series of subgroup and moderator analyses. These analyses fall into three categories: 1) analyses regarding theoretically relevant categorical moderators, 2) analyses of methodological and bias-related categorical moderators, and 3) analyses concerning theoretically relevant continuous moderators. For the first and second sets of analyses, we used the partially empirical subgroup correlated-effects plus (PESCE+[RVE]) model[63]. For the third category, we relied on the PECHE-RVE model, as presented in

Equation (5). To examine whether mental health and social reintegration effects tend to covary, we fitted the partially correlated and multivariate effects plus model (PECVME; see Pustejovsky & Tipton[63], first version on OSF).

Specifically, we applied the PESCE+ model combined with RVE and cluster wild bootstrapping when investigating binary or categorical moderators. Generally, this amounts to conducting individual PECHE meta-analyses[6] across each subgroup while accounting for studies contributing $k_j \geq 1$ effects to $C \geq 1$ subgroups (i.e., the model handles dependent effect sizes). Importantly, this allows us to estimate reliable Wald test *p-values* with adequate statistical properties. To test whether single-subgroup effects were statistically distinct from null, we used RVE (i.e., the HTZ test), whereas we used cluster wild bootstrapping to estimate Wald test p-values comparing differences between the subgroup effects.[7] In line with the estimation of the overall average effect size, we fitted this model to include heterogeneity at both the effect and study levels (indicated by the + sign).

List of moderators

As per protocol, the moderator analyses included the following factors:

Theoretical factors:

- The measured outcome (e.g., alcholhol and drug use, social functioning, well-being and quality of life)
- The sample characteristics including:
    - samples with vs. without participants with schizophrenia,
    - the average percentage og males in the sample,
    - the average age of the participants.

- The type of intervention (cognitive-behavioral therapy [CBT] vs. other types of interventions).
- The number of intervention sessions
- The duration of the intervention

[6] Although the weighting scheme slightly varies across the SCE and CHE models (see Pustejovsky, 2020b; Supplementary material in Pustejovsky & Tipton, 2022). However, the heterogeneity measures would be the same as if one fitted the CHE model on each subgroup category.

[7] This follows the recommendation from Joshi, Pustejovsky et al. (2022), which is: "we recommend using CWB rather than HTZ for tests of multiple-contrast hypotheses in meta-analyses conducted using RVE. Tests of single-contrast hypotheses (i.e., t-tests) can be conducted using either the CWB or HTZ test because both methods have very similar power." (p. 473)

Methodological factors

- The preregistration status
- The type of test (clinician-rate vs. self-reported)
- The analytical strategy (ITT vs. TOT)
- The research design ([C]RCT vs. QES)
- The type of control group (individual treatment vs. treatment as usual (TAU and waitlist)
- The overall risk of bias status (high vs. low/moderate overall risk of bias
- The timing of the measurement after the end of the intervention.

Across all of the above-listed meta-regression analyses, we fitted models with and without controlling for other covariates. In the covariate-adjusted models, we controlled for 1) the type of outcome, 2) whether the sample included participants with schizophrenia, 3) whether the group-based intervention was based on CBT, 4) preregistration status, 5) type og test, 6) the analytical strategy, 7) the research design ([C]RCT vs. QES), 8) type of control group, 9) the risk of bias rating, 10) average age of study sample, 11) percentage of males in sample, 12) number of session per week, 13) length of intervention (duration), and finally 14) measurement timing. To guard against multicollinearity, we only added covariates that did not show intercorrelations above 0.5. Consequently, we did not add the percentage of males and the number of sessions indicators, as these factors were highly correlated with the sample composition and duration, respectively, for social reintegration outcomes. For models including mental health outcomes, we did not add the risk of bias, control group, and number of sessions indicators to the models, as these factors were highly correlated with the research design, age, and duration, respectively. Find the full correlation matrix used for the selection of variables in the Appendix B Tables 36 and 41.

To facilitate and ensure the correct interpretation of overall average subgroup means from the PESCE+ model[72], we centered all binary, categorical, and continuous control variables. Binary variables were mean-centered. For categorical variables, we created dummy variables for each category and mean/proportion-centered each dummy variable. Continuous variables were centered using rounded median-centering. When controlling for outcome-measure type, we added a separate mean-centered dummy variable for each outcome measure to the model.

Further model details and notation for PESCE+

The PESCE+[RVE] model accounts for dependent effect sizes by assuming that effects from the same studies falling into the same subgroup are correlated, whereas effects falling into different subgroups are assumed to be uncorrelated. Despite more advanced models being available that account for correlations between effects from the same study appearing across different subgroups, these models have only been shown to perform adequately under specific data structures, which were only present at the most generic level of the data. Therefore, we used the PESCE+[RVE], which formally takes the following form

$$T_{ij} = \sum_{c=1}^{C} m_{ij}^{c}\left(x_{ij}\boldsymbol{\beta_c} + u_{cj} + v_{cij}\right) + e_{ij} \tag{6}$$

Here, m_{ij}^{c} $\left(m_{ij}^{1} \dots m_{ij}^{C}\right)$ is an indicator of whether the effect size i from study j falls in subgroup c for all $c = 1 \dots C$. If so, $m_{ij}^{c} = 1$, otherwise $m_{ij}^{c} = 0$. The model is based in the following assumptions: $u_{cj} \sim N(0, \tau_c^2)$, $v_{cij} \sim N(0, \omega_c^2)$, and $e_{ij} \sim N(0, s_{ij}^2)$, $Cov\left(u_{cj}, u_{bj}\right) = 0$, $Cov\left(v_{cij}, v_{bij}\right) = 0$, and $Cov\left(e_{ij},\ e_{lj}\right) = \rho s_{ij} s_{lj} \sum_{c=1}^{C} m_{ij}^{c} m_{lj}^{c}$, thereby $Cov\left(e_{ij},\ e_{hj}\right) = 0$ when $m_{ij}^{c} \neq m_{hj}^{c}$.

Examination of covariation between social reintegration and mental health outcomes

As the majority of studies both reported on social reintegration and mental health outcomes, we used, as mentioned above, the PECVME model to statistically examine whether group-based interventions with larger impacts on social reintegration outcomes also tended to have greater impacts on mental health outcomes. The major difference between PECMVE and PESCE is that the PECMVE model allows effects from the same studies falling in different subgroups to be correlated. The model takes the following form

$$T_{ij} = x_{ij}\boldsymbol{\beta} \sum_{c=1}^{C}\left(m_{ij}^{c} u_{cj} + m_{ij}^{c} v_{cij}\right) + e_{ij} \tag{7}$$

where $u_{cj} \sim N(0, \tau_c^2)$, $v_{cij} \sim N(0, \omega_c^2)$, and $e_{ij} \sim N(0, s_{ij}^2)$, $Cov\left(u_{cj}, u_{bj}\right) = \psi_{bc}\tau_b\tau_c$, $Cov\left(v_{cij}, v_{bij}\right) = \zeta_{bc}\omega_b\omega_c$, and $Cov\left(e_{ij},\ e_{lj}\right) = \rho s_{ij} s_{lj}$. ψ_{bc} and ζ_{bc} are the correlations between the random effects at the study and effect size level, respectively.

Sensitivity analysis

To test the robustness of our results, we conducted a range of sensitivity analyses. First, we tested if our results were sensitive to the assumed correlation among within-study effects, as this assumption can impact some parts of the model estimation. Specifically, we reestimated all results by setting $\rho = 0,\ .2,\ .4,\ .6.,\ .9$. Note, the first specification amounts to fitting a classical multi-level meta-analysis model[73,74] (MLMA-RVE).

Second, we ran sensitivity analyses across different types of effect size metrics. That is, we reestimated all results using a covariate/pretest-adjusted version of both Cohen's d and Hedges' g (i.e., without adjusting for clustering), g_t (c.f. Equation 1) standardized by its original standard deviation and not the population-based version, and a posttest-only version of g_t. We also tested the sensitivity across different assumptions of the interclass correlation (ICC) used to cluster-bias-adjust effect sizes, setting $ICC = 0.05, 0.2$.

Finally, we investigated whether the impact of removing high-risk of bias effect and non-randomized studies on our results. We did not examine the impact of outliers, as we did not find any one as defined as effect sizes falling more than three times the interquartile range below the first quartile or above the third quartile[28,64], see Figures S7 and S8.

Assessment of reporting biases

We conducted a range of complementary publication bias and/or small study effects tests (henceforth publication bias tests[8]), as no publication bias test clearly outperforms all other methods. Also, this follows the general recommendations for publication bias testing in meta-analysis[76–79]. Specifically, we conducted the following tests separately for the social reintegration (primary outcome) and mental health (secondary outcome) data:

1. Bootstrap versions of the newly developed hybrid extended meta-analysis (HYEMA) tests[80]
2. Worst-case sensitivity analysis tests[81]
3. p-uniform* tests[82]
4. Robust and adjusted version of the partially empirically correlated-hierarchical effects model that incorporates inverse sampling covariance weights (PECHE-ISCW)[77,83]
5. Robust and adjusted versions of PET/PEESE, incorporating ISCW[77]
6. The newly developed bootstrapped step-function selection models for meta-analysis of dependent effect sizes[60]

[8] We acknowledge that small study, reporting, and publication bias tests are not the same and that many statistical tests cannot distinguish between these types of bias. For simplicity, however, we subsumed all these tests under the heading of publication bias tests, similar to Rothstein et al.[75].

On the one hand, we chose these tests because they have shown the most promising statistical properties in the simulation studies embedded in the cited studies above, as well as shown in independent evaluations of publication bias methods[76,79]. On the other hand, Chen and Pustejovsky[77] and Pustejovsky, Citkowicz et al.[60] have shown that regression-based methods (such as PECHE-RVE-ISCW and PET/PEESE-RVE-ISCW) are generally better relative to selection models at detecting publication biases when selection is weak, whereas the selection models clearly outperform regression-based methods in the presence of moderate to strong selection in effect sizes. We, therefore, aimed to include a mixture of publication bias tests that function well under different levels of selective reporting.

To differentiate between publication bias and systematic and substantial differences between effect sizes, we conducted two types of tests, where we adjusted subgroup effects for publication bias. Specifically, we adjusted subgroup effects for publication bias for preregistration status and outcome type. As not all publication bias tests can be used for correcting subgroup effects, we only used a subset of the above-listed tests. Find information on the exact models used for this type of publication bias adjustment in the next section.

Across all models, despite the worst-case meta-analysis models, we used a modified version of the standard error and variance presented in Equation (3). More precisely, we defined the modified version of the standard error and variance as follows:

$$V_{gt}^{mod} = W \times \xi \text{ and } SE_{mod} = \sqrt{V_{gt}^{mod}}. \tag{4}$$

As described by Pustejovsky and Rodgers[37], this approach is necessary to avoid the artificial correlation between SMD effect size estimates and their sampling variance, which arises because SMD estimates are used to calculate P in Equation (3). For a better understanding of the relative difference between V_{gt}^{mod} and V_{gt}, see Figures 98 and 99 in Appendix B. We did not use modified standard errors in the worst-case meta-analysis, as it is intended to represent a sensitivity analysis that replicates the main analysis, just without including effect sizes that affirm one's non-null hypothesis and are statistically significant.

Visualization

For visualization purposes, we applied contour-enhanced funnel plots[84], which aim to depict the relationship between the effect sizes and their estimated standard errors. In these plots, we included the estimated slope from the robust Egger's regression tests[80], and colored the effect size estimate by their overall risk of bias assessment. We both visualize funnel plots of the study and effect size levels. For the study-level plots, we averaged[9] all within-study effect sizes, assuming a constant between-effects correlation of 0.8. As all of our risk of bias assessments were conducted at the effect size level, we did not color the average effect sizes in study-level plots. Find funnel plots including study-level assessments in the Online Appendix E.

Selective reporting and preregistration

A special feature of our data is that approximately half of the included studies (22 of 46 studies in the social integrational data and 20 out of 42 studies in the mental health data) were preregistered. In preregistered studies, one could expect that publication bias is either completely absent or at least much less pronounced compared to conventional, non-preregistered studies. For this type of effect size data, it has recently been suggested not to correct preregistered studies for publication bias[80] or to model this factor[60]. In line with these recommendations, we assessed publication bias by only adjusting non-preregistered studies or by adding a centered dummy variable for preregistration status to the given publication bias model. Centering of binary variables follows the recommendation forwarded by Fisher and Tipton[72]. Moreover, we depicted funnel plots separately for preregistered and non-preregistered studies.

Model-specific details for publication bias tests

Cluster bootstrap HYEMA

As the only test, HYEMA provides average effect size estimates where only effect sizes from non-preregistered studies are adjusted for publication bias. Although this test has shown promising performance[80], it has only been evaluated under the assumption of independence among effect sizes (i.e., assuming all studies contribute one effect size only). To overcome this issue and to control the nominal Type I error rate, we cluster-boostrapped this model[59]. For the bootstrap models, we calculated the percentile confidence intervals, as these have shown the most promising performance in other applications[60].

[9] We used the aggregate.escalc() function from the metafor package for the aggregation.

Furthermore, we used the HYEMA model to adjust outcome moderator effects for publication bias. For multi-constrast tests, investigating whether effect sizes differed across different types of outcome, we computed bootstrap Wald test p-values defined as

$$p = \frac{1}{R}\sum_{r=1}^{R} I(F^r > F)$$

Where R is the total number of cluster bootstrap replications, and F is the naïve F-test. We used similar formulas to calculate the naïve F-test, similar to those implemented in the *rma*() function in metafor. Across all of these tests, we used $R = 1999$.

*p-uniform**

The p-uniform* method intends to adjust for publication bias by estimating the overall average effect size that makes the distribution of p-values in the data as uniform as possible. Apparently, a downside of the p-uniform* method is that it is based on the assumption of independence among effect sizes, which by design makes it miscalibrated when applied with dependent effect sizes. Nonetheless, Chen and Pustejovsky[77] showed that the method performs well even in dependent effect size data, which is the main reason why we included this method. Yet, this method has only been developed to adjust the overall average mean effect, and we therefore only used it for this purpose.

Worst-case meta-analysis

The worst-case meta-analysis is a sensitivity analysis in which all positive and statistically significant effect sizes were excluded, under the extreme assumption that all represent false positives. If the overall average effect remains statistically significant and substantial in size, this provides strong evidence that publication bias is not the primary factor driving the effect(s). We used this type of analysis to re-estimate the overall average effect size as well as subgroup effects across preregistration status and types of outcome. For the latter analysis, we controlled for preregistration status.

Partly to ease comparison between publication bias tests and partly to ease the presentation of these tests, we only fitted these models using the PECHE-RVE model as presented in Equation (5) above[63,85]. For the subgroup test, this model type slightly differs from the one used in the main analysis. To align the models, we also fitted moderator models using the same model as in the main subgroup analyses. These tests can be found in the publication bias script in the Online Appendix E.

(PE)CHE-RVE-ISCW

Next, we used the newly developed PECHE-RVE-ISCW model to test for publication bias[77]. The key feature of this model relative to the original CHE-RVE model (see Equation 5) is that it incorporates weights that are based on inverse sampling variance estimates and the assumed covariance between these estimates. When using the modified version of the variance described in Equation (4), this corresponds to fixed-effect weighting, or more precisely, weighting by the inverse of the effective sample sizes and the assumed covariance among these.

To guard against model mis-specification and adjust for small sample issues, we used RVE, or more precisely, the robust HTZ for single-contrast tests[86]. This type of test was used for the overall average effect size and single-subgroup effects estimations, following the recommendation by Joshi, Pustejovsky et al.[58]. When estimating multiple-contrast hypothesis tests, we estimated cluster wild bootstrap Wald test *p* values[58].

We used this test to adjust for publication bias in both the overall average effect size and the moderator effects across preregistration status and outcome types

PET/PEESE-RVE-ISCW

The PET/PEESE-RVE-ISCW models we used had the same shape as the PECHE-RVE-ISCW model, with the only exception that we either added the modified sampling variance (PEESE; precision-effect estimator with standard error) or the modified standard error (PET; precision-effect test) presented in Equation (4) as a predictor in the models. As with the PECHE-CHE-ISCW model, we used this test to adjust the overall average effect size as well as the moderator effects across preregistration status and types of outcome. Differently, and in addition to the PECHE-CHE-ISCW model, we also controlled for the preregistration status when estimating the overall average effect.

It has been recommended by Stanley and Doucouliagos[87] that "[w]hen the PET test is not rejected, meaning that the average effect is not statistically distinguishable from zero, then the PET intercept is used for estimating the adjusted average effect. However, if the PET test is rejected and the average effect is statistically distinct from zero, the PEESE is used for estimating the adjusted average effect"[77]. We followed this decision rule when re-estimating the overall average effect with the PET/PEESE models. For subgroup models, we used the PET-RVE-ISCW only, as the decision rule was not clearly defined in this context.

Cluster bootstrap selection models

As selection models have shown promise across a range of simulations[60,76,77,83], we applied two versions of these types of models. That is, we used the three-parameter selection model (3PSM) and the four-parameter selection model (4PSM). The former included a single parameter, describing the "likelihood of non-affirmative effect sizes being observed relative to affirmative effect sizes."[77]. For this model, we set the step parameter to 0.025, amounting to the threshold of a classical one-sided *p*-value. The latter model included two-step parameters, which we set to .025 and 0.50. This allowed for modeling different probabilities for non-significant effect sizes, conditional on the direction of the effect.

To account for dependencies in our effect size data, we used cluster bootstrap selection models[60]. Specifically, we fitted these models using the composite marginal likelihood (CML) estimator with two-stage bootstrapped percentile confidence intervals based on 1999 re-sampled replicates, as this model has shown the best performance relative to other estimators and confidence intervals (see Figures 4 & 5 in Pustejovsky, Citkowicz, et al.[60]).

As with the PECHE-RVE-ISWC model, we used this test to adjust for publication bias in both the overall average effect size and the moderator effects across preregistration status and outcome types. In the outcome model, we controlled for the preregistration status.

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